



Private Higher School of Engineering and Technology
ESPRIT School of Engineering

TECHNICAL REPORT

Digital Solution for Scout Group Management

Client : Scout Group — Grombelia Al-Medina (Tunisian Scouts)

Developed By : BISMART

Supervised By : Ridha Berrahal

Malek NAGHMOUCHI

Team Members : Eya Rhouma

Mortadha Houimeli

Mohamed Dhia Chaaben

Salma Korbi

Mohamed Dhia Eddine Brahmi

Aya Louhichi

Acknowledgements

We would like to express our sincere gratitude to all those who contributed to the successful completion of this project.

First and foremost, we extend our thanks to the **Scout Group of the City of Grombelia (Fawj Grombelia Al-Medina)** and the **Tunisian Scout Federation** for their trust, their availability, and for providing the data and organizational context that made this project possible. Their openness to innovation and data-driven governance was a constant source of motivation throughout our work.

We are deeply grateful to our **academic supervisors and coaches** for their guidance, their rigorous technical reviews at each sprint milestone, and their invaluable feedback that continuously improved the quality of our deliverables. Their expertise in Business Intelligence, Machine Learning, and software engineering shaped both the methodology and the implementation of this platform.

Our appreciation extends to the **stakeholders** who participated in validation sessions, provided domain knowledge, and helped us translate organizational needs into actionable analytical requirements. Their input was essential in grounding every technical decision in real business value.

Finally, we thank our **teammates and peers** for the collaborative spirit, the long working sessions, and the shared commitment to delivering a solution that genuinely serves its users. This project is the result of collective effort, mutual support, and a shared passion for data and technology.

The Project Team
BISMART

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List of Abbreviations and Acronyms

AI : Artificial Intelligence
API : Application Programming Interface
ARIMA : AutoRegressive Integrated Moving Average
BI : Business Intelligence
CSV : Comma-Separated Values
CRISP-DM : Cross-Industry Standard Process for Data Mining
DAG : Directed Acyclic Graph
DBMS : Database Management System
DBSCAN : Density-Based Spatial Clustering of Applications with Noise
DWH : Data Warehouse
EDA : Exploratory Data Analysis
ETL : Extract, Transform, Load
FR/EN : French / English
GMM : Gaussian Mixture Model
IQR : Interquartile Range
JSON : JavaScript Object Notation
JWT : JSON Web Token
KPI : Key Performance Indicator
LLM : Large Language Model
LSTM : Long Short-Term Memory
MAE : Mean Absolute Error
MAPE : Mean Absolute Percentage Error
ML : Machine Learning
MLOps : Machine Learning Operations
MSE : Mean Squared Error
PCA : Principal Component Analysis
RMSE : Root Mean Squared Error
REST : Representational State Transfer
SCD : Slowly Changing Dimension
SPA : Single Page Application
SQL : Structured Query Language
SVD : Singular Value Decomposition
UI : User Interface

1 Executive Summary

This report presents the full project lifecycle of a Business Intelligence, Machine Learning, and Artificial Intelligence solution developed for the Scout Group of the City of Qarnbaliya — *al-Madina Qarnbaliya Fawj* (Tunisian Scouts). The project follows the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) methodology, covering six iterative phases from business understanding through deployment.

The client manages a significant volume of data per scout season — including member registrations, unit-level activities, budgets, and leadership structures. Data was previously exploited only for administrative tracking. This project transforms that fragmented data landscape into a unified, intelligent decision-support system.

Key deliverables include:

- A fully automated ETL pipeline built with **Talend**, orchestrated via **Apache Airflow**
- A star-schema **Data Warehouse** populated with fact and dimension tables
- **Web scraping** pipelines to enrich internal data with external sources
- **Power BI** dashboards with French/English bilingual support, embedded in an **Angular** web application
- Machine learning models: clustering, anomaly detection, recommendation systems, regression, and time-series forecasting
- A **Flask REST API** serving ML model predictions
- **Facial recognition** and password-based authentication
- An AI-powered **chatbot** for interactive decision support
- Real-time infrastructure monitoring via **Grafana**
- MLOps lifecycle management using **LMFlow**

2 Phase 1 — Business Understanding

Phase 1 Business Understanding

Introduction

In this phase, we focused on understanding the organizational context and the business needs of the Scout Group of Grombelia Al-Medina. We identified the main stakeholders, defined the project objectives, analyzed the existing management limitations, and established the functional scope of the future BI/ML/AI platform.

This phase allowed us to align the technical vision of the project with the real operational and strategic expectations of the scout organization.

2.1 Background and Context

The Scout Group of Grombelia (*Fawj Grombelia Al-Medina*) operates under the Tunisian Scout Federation and manages a structured hierarchy of scout units differentiated by age group, gender, and activity type. Each scout season generates rich longitudinal data on membership evolution, activity participation, budget allocation, and leadership capacity.

Historically, this data was captured in isolated administrative records with no unified analytical layer. Decision-makers lacked the tools to identify membership trends, compare unit performance, or anticipate resource requirements. This project bridges that gap through a fully integrated BI/ML/AI platform.

2.2 Business Objectives

The primary business objective is to place data at the service of governance and strategic planning for the scout group. Specific objectives include:

- Centralizing and structuring information to provide a global, coherent vision across seasons and units
- Improving the quality and speed of decision-making for group leadership
- Understanding past and current membership, activity, and budget trends
- Anticipating future evolutions through predictive analytics
- Introducing AI mechanisms to support organizational decision-makers

2.3 Functional Scope

Table 1. Functional Scope by Domain

Domain	Functional Coverage
Members	Registrations, demographic breakdown, seasonal evolution of headcount
Scout Units	Activity levels, unit performance metrics, leadership ratios
Activities	Meetings, outings, camps, exceptional events, participation rates
Budgets	Forecasts, actual spend, resource consumption per activity/unit
Temporal Analysis	Cross-season comparison, trend identification
Predictive & AI	Forecasting, clustering, anomaly detection, recommendations

2.4 Key Performance Indicators (KPIs)

Table 2. Key Performance Indicators

KPI	Target / Method	Tool
Total members	Count by unit/season	Power BI / DWH
Membership evolution rate	Season-over-season % change	Power BI / ML Forecast
Activity participation rate	Participants / Total members	Fact_Participation_Activity
Avg. activities per unit	Count per unit per season	Power BI
Total budget per season	SUM(amount) by season	Fact_Budget
Budget consumption rate	Actual / Planned budget %	Power BI
Cost per member	Budget / nb_members	Fact_Budget + Fact_Membership
Member retention rate	Returning members ratio	Fact_Membership
Units at risk	Predicted low-engagement clusters	ML Clustering

2.5 Stakeholder Analysis

Three primary stakeholder groups were identified and validated:

- **Group Leadership (Chef de Groupe):** Strategic vision, budget oversight, season planning
- **Unit Leaders (Chefs d'Unité):** Operational activity management, member engagement tracking
- **Treasurer:** Registration management, financial record-keeping

Conclusion

In this chapter, we presented the business understanding phase of the project. We analyzed the organizational context, identified the stakeholders, defined the business objectives, and established the key performance indicators that will guide the analytical process.

These elements provided a solid foundation for the continuation of the project. In the next chapter, we will focus on understanding the available data sources and analyzing the structure and quality of the collected data.

3 Phase 2 — Data Understanding

Phase 2

Data Understanding

Introduction

In this chapter, we focus on the data understanding phase. We present the different internal and external data sources used in the project, analyze the structure of the data warehouse schema, and perform an exploratory analysis of the collected information. The objective of this phase is to evaluate the quality, consistency, and analytical potential of the available data before starting the preparation and transformation processes.

3.1 Data Sources

Multiple data sources were identified and ingested during the project:

Table 3. Data Sources and Ingestion Methods

Source		Description & Method
Internal Records	Scout	Historical Excel/CSV exports of member registrations, unit structure, activity logs, and budget records
Web Scraping		External data collected via Python (BeautifulSoup / Scrapy) to enrich activity and geographic context
Reference Data		Manually curated unit configurations, age thresholds, gender classification, and season definitions

3.2 Data Schema — Schema

The Data Warehouse follows a star schema architecture. Figure 1 illustrates the conceptual diagram of the schema.

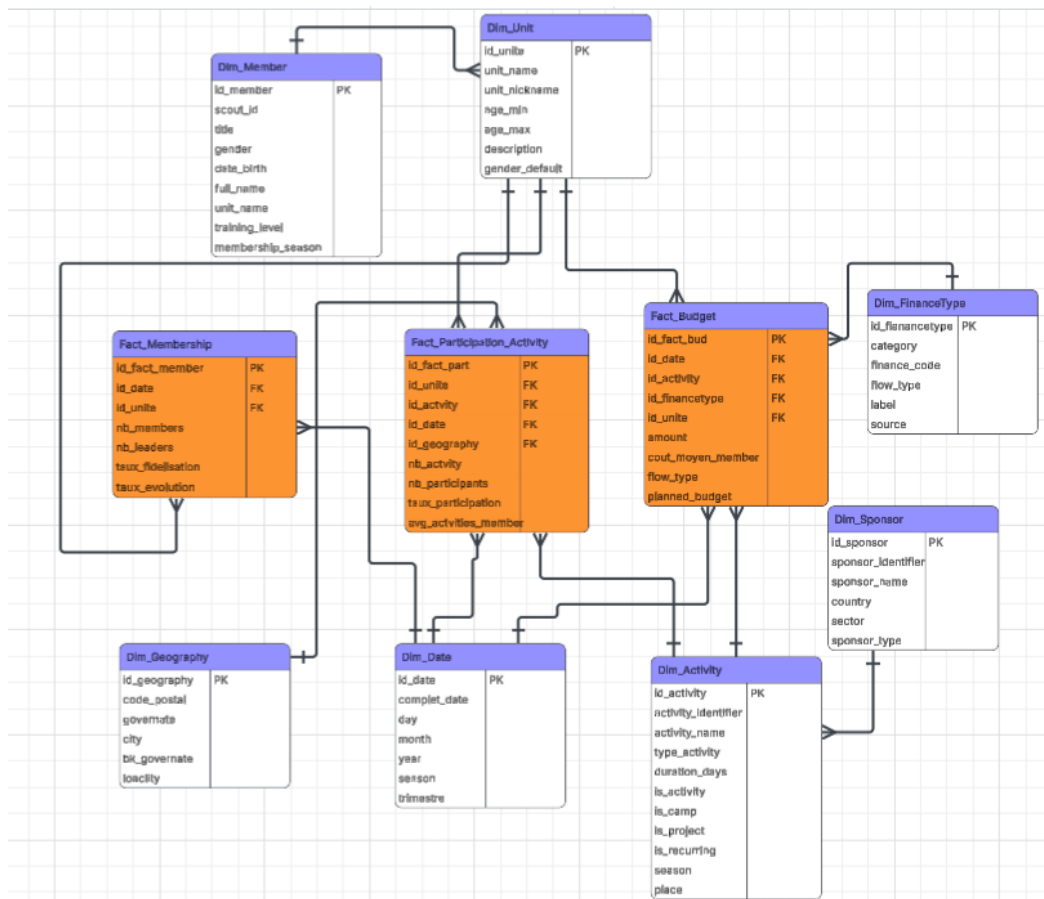


Figure 1. Data Warehouse Schema

3.2.1 Fact Tables

- **Fact_Membership:** id_fact_member, id_date, id_unite, nb_members, nb_leaders, taux_fidelisation, taux_evolution
- **Fact_Participation_Activity:** id_fact_part, id_unite, id_activity, id_date, id_geography, nb_activity, nb_participants, taux_participation, avg_activities member
- **Fact_Budget:** id_fact_bud, id_date, id_activity, id_financetype, id_unite, amount, cout_moyen_member, flow_type, planned_budget

3.2.2 Dimension Tables

- **Dim_Member:** id_member, scout_id, title, gender, date_birth, full_name, unit_name, training_level, membership_season
- **Dim_Unit:** id_unite, unit_name, unit_nickname, age_min, age_max, description, gender_default
- **Dim_Date:** id_date, complet_date, day, month, year, season, trimestre
- **Dim_Activity:** id_activity, activity_identifier, activity_name, type_activity, duration_days, is_activity, is_camp, is_project, is_recurring, season, place

- **Dim_Geography:** id_geography, code_postal, governate, city, blk_governate, locality
- **Dim_FinanceType:** id_financetype, category, finance_code, flow_type, label, source
- **Dim_Sponsor:** id_sponsor, sponsor_identifier, sponsor_name, country, sector, sponsor_type

3.3 Initial EDA Findings

Exploratory Data Analysis (EDA) was conducted via Jupyter notebooks and revealed the following:

- Significant seasonal variance in membership counts across units, with certain units experiencing consistent decline
- Activity participation rates showed strong correlation with unit size and leadership ratio
- Budget consumption was highly uneven — larger units consumed disproportionate resources relative to headcount
- Missing values identified in geographic fields and membership season assignments
- Outlier detection flagged multiple budget entries requiring validation prior to modeling

3.4 Data Quality Assessment

Table 4. Data Quality Issues and Resolutions

Issue Identified	Resolution Applied
Inconsistent season labeling	Standardized using Dim_Date season field
Duplicate member entries	Deduplication using scout_id as natural key
Outlier budget entries	Flagged via IQR method; verified with client
Inconsistent activity type labels	Mapped to controlled vocabulary in Dim_Activity

Conclusion

In this chapter, we explored the different data sources and analyzed the structure of the data warehouse schema.

We also conducted an exploratory data analysis and identified several data quality issues that required preprocessing and normalization.

The obtained insights helped us better understand the characteristics of the dataset

and prepare it for the next phase. In the following chapter, we will focus on data preparation, ETL processing, and pipeline orchestration.

4 Phase 3 — Data Preparation

Phase 3 Data Preparation

Introduction

This chapter presents the data preparation phase of the project. We describe the staging architecture, the ETL workflows implemented using Talend, the integration of external data through web scraping, and the orchestration of the entire pipeline using Apache Airflow.

The goal of this phase is to transform raw and heterogeneous data into a clean, structured, and reliable analytical dataset ready for machine learning and business intelligence processes.

4.1 Staging Area

A dedicated internal staging area was established as the first integration layer. Raw source files were deposited here before any transformation, preserving original data integrity and enabling reprocessing if needed.

4.2 ETL Pipeline — Talend

The ETL process was developed using **Talend Open Studio for Data Integration** and covers three phases:

4.2.1 Extract

- Connection to flat-file sources (CSV, Excel) and staged data
- Web scraping outputs ingested via flat-file connectors
- Schema validation at point of extraction

4.2.2 Transform

- Data type normalization and date standardization
- Derivation of calculated fields: `taux_fidelisation`, `taux_evolution`, `taux_participation`, `avg_activities_member`, `cout_moyen_member`
- Surrogate key generation for all dimension tables
- Slowly Changing Dimension (SCD Type 1) handling for `Dim_Member` and `Dim_Unit`
- Lookup joins to resolve foreign keys between fact and dimension layers
- Filter logic to exclude invalid or test records

4.2.3 Load

- Incremental load strategy to the target Data Warehouse
- Referential integrity enforcement during load

- Error rows redirected to rejection tables for audit

4.3 Web Scraping

Python scripts were developed to supplement internal data with externally sourced information:

- **Geographic enrichment:** governorate and city data scraped from public municipal databases
- **Activity context:** outdoor and scouting event schedules from public Tunisian federation pages
- **Sponsor information:** partner data from publicly accessible directories

Scraped data was cleaned and integrated into the staging area before ETL loading.

4.4 Orchestration — Apache Airflow

The entire data pipeline was orchestrated using **Apache Airflow** with DAGs (Directed Acyclic Graphs) defined for:

- Daily staging refresh jobs
- Weekly full ETL runs to the Data Warehouse
- ML feature extraction pipelines triggered post-DWH update
- Alert mechanisms for pipeline failures and data quality thresholds

Conclusion

In this chapter, we presented the complete data preparation workflow, including staging, extraction, transformation, loading, and orchestration processes. We also integrated external data sources through web scraping and automated the pipelines using Apache Airflow.

The prepared and validated data now constitutes a robust analytical foundation for the machine learning and artificial intelligence models developed in the next chapter.

5 Phase 4 — Modeling: Machine Learning & AI

Phase 4 Modeling — Machine Learning & AI

Introduction

In this chapter, we present the modeling phase of the project, which represents the core analytical component of the platform. Several machine learning and artificial intelligence models were designed, implemented, and evaluated in order to support predictive analytics, anomaly detection, recommendation generation, clustering, and decision support.

Different approaches were compared to identify the most suitable models according to the characteristics of the scout organization data and the project objectives.

5.1 Recommendation System

5.1.1 Objective

The recommendation system was designed to automatically suggest relevant activities to scout members based on their historical participation patterns and the behavioral similarities observed across other members and units. Rather than relying on manual scheduling or leadership intuition alone, this module provides a data-driven layer of activity planning that personalizes suggestions at the individual member level.

5.1.2 Chosen Approach — Collaborative Filtering (SVD / Matrix Factorization)

Collaborative Filtering was selected as the core methodology, specifically the **Singular Value Decomposition (SVD)** technique applied to Matrix Factorization. This approach operates on the premise that members who have historically participated in similar activities share underlying preference profiles, and that these latent patterns can be used to predict unobserved participation scores. SVD decomposes the user-activity interaction matrix into lower-dimensional latent factors, enabling the model to generalize from sparse observed data and recover missing preference scores with high fidelity.

5.1.3 Implementation

Three tables from the Data Warehouse were used as input: `Dim_Activity` (the activity catalog), `Dim_Member` (the member registry), and `Fact_Participation_Activity` (participation records). A user $\times$ activity matrix was constructed using `nb_participants` as a proxy for the implicit rating signal. The SVD decomposition was then applied to learn the latent representations of both users and activities. From these reconstructed preference scores, Top- N recommendations were generated for each member, ranking activities by their predicted participation likelihood.

5.1.4 Evaluation

Reconstruction Metrics — SVD Matrix Factorization

• RMSE $\approx 4.42 \times 10^{-14}$

• MAE $\approx 1.65 \times 10^{-14}$

These near-zero error values confirm that the SVD decomposition reconstructed the participation matrix with exceptional fidelity, validating the choice of algorithm for this dataset structure.

5.1.5 Results

As an illustrative example, Top-*N* recommendations generated for User 1 yielded:

- **Activity 17** — *Summer Camp Cubs Flowers 2025* → predicted score ≈ 85
- **Activity 49** — *General Activities* → predicted score ≈ 36
- All remaining activities returned scores close to 0, reflecting either low historical participation or an absence of behavioral signal.

A bar chart visualization of Top-*N* Recommended Activities was produced for each member (Figure 2), clearly illustrating the differentiation in predicted preference scores.

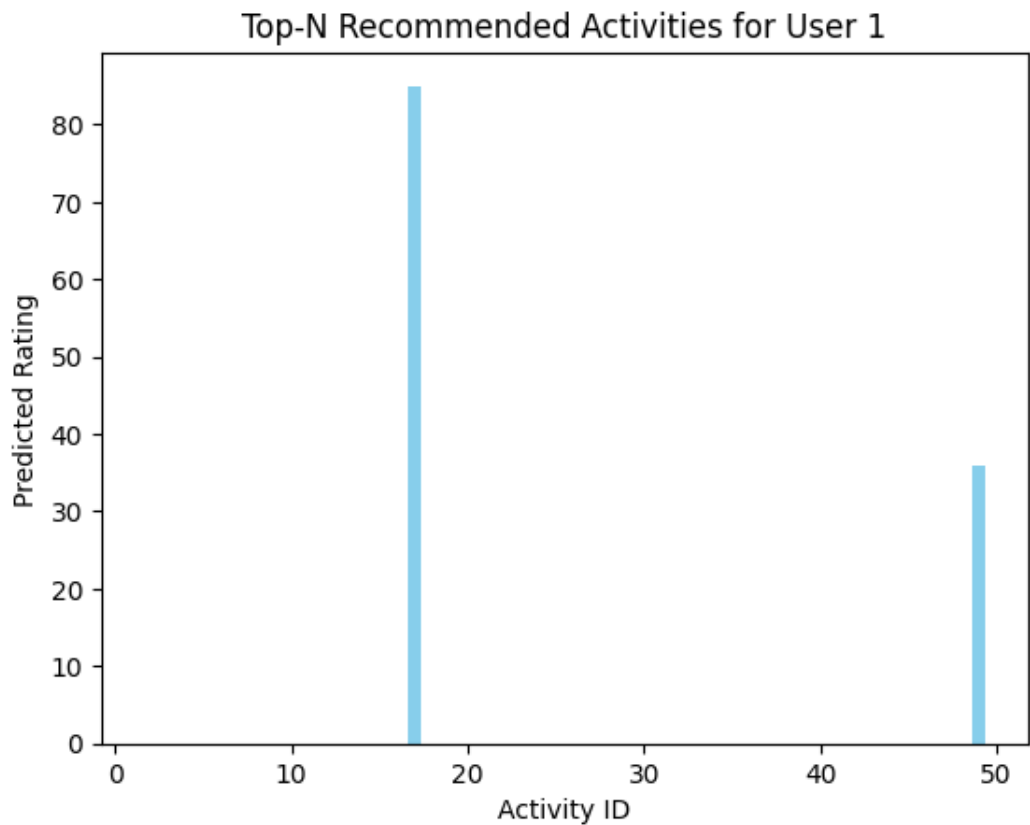


Figure 2. Top-N Recommended Activities for User 1 (SVD Score)

5.2 Anomaly Detection

5.2.1 Objective

The anomaly detection module was developed to identify atypical values within the budget data, with the aim of detecting data entry errors, potential financial irregu-

larities, and exceptional events that deviate from established organizational patterns. Early identification of such anomalies allows group leadership to investigate and respond before issues propagate into downstream reporting or planning.

5.2.2 Methods

Two complementary approaches were implemented and compared:

- **Isolation Forest** isolates anomalous observations by randomly partitioning the feature space. Anomalies — being statistically rare and structurally distant from the bulk of the data — require fewer random splits to isolate, resulting in lower anomaly scores. This algorithm is particularly well-suited to heterogeneous financial records and does not assume a specific distributional form.
- **Z-Score and IQR (Interquartile Range)** methods provided a statistical baseline. The Z-Score approach flags observations beyond a defined number of standard deviations from the mean; the IQR method flags values outside $1.5\times$ the interquartile range. These simpler methods served as a validation layer, confirming anomalies detected by Isolation Forest against classical statistical criteria.

Results were visualized through a Budget vs. Season scatter plot (Figure 3) in which detected anomalies were highlighted in red, providing an immediately interpretable view for non-technical stakeholders.

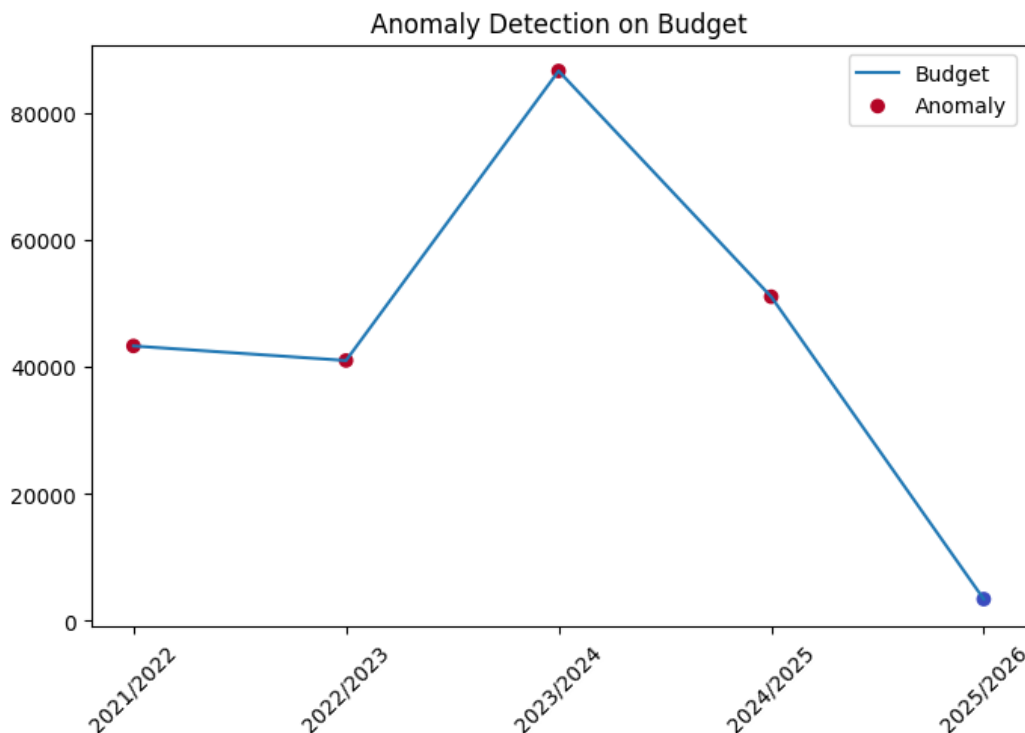


Figure 3. Budget Anomaly Detection — Budget vs. Season

5.3 Time Series Forecasting

5.3.1 Objective

The forecasting module was designed to predict the future evolution of key organizational indicators — including membership headcount, budget amounts, and participa-

tion rates — in order to anticipate trends and provide an empirical basis for strategic season planning.

5.3.2 Stationarity Analysis

Prior to model selection, the time series data was subjected to stationarity testing:

- The **Augmented Dickey-Fuller (ADF)** test returned a high p-value, indicating that the raw series was non-stationary.
- The **KPSS** test confirmed the presence of a deterministic trend.

Decomposition analysis revealed a clear long-term trend, no significant seasonal component, and low residual variance — suggesting parsimonious models would be appropriate.

5.3.3 Models Implemented

- **ARIMA (1,1,1)**: A classical time series model combining autoregression, differencing, and a moving average component. Its differencing order of 1 directly addresses the non-stationarity identified by the ADF test, making it a theoretically justified choice. ARIMA is particularly robust for short series with limited historical depth.
- **Prophet (Meta)**: A flexible additive forecasting model designed to handle trend changes, seasonality, and missing data gracefully. Well suited to producing interpretable multi-step projections with uncertainty intervals.

5.3.4 Evaluation

Table 5. Time Series Model Evaluation — MAE, RMSE, MAPE

Model	MAE	RMSE	MAPE
ARIMA (1,1,1)	5.15	5.15	42.95%
	9.49	9.49	79.09%

ARIMA outperformed Prophet across all three metrics. Its superior performance is consistent with the characteristics of the data: a short series with a clear trend but no strong seasonal structure.

5.3.5 Forecast Results

Budget Projections — Future Scout Seasons

- Season 2024/2025 → ≈ 72,146
- Season 2025/2026 → ≈ 78,394
- Season 2026/2027 → ≈ 84,432

Prophet independently confirmed an upward trend across seasons 2026, 2027, and 2028, lending cross-model credibility to the projected growth trajectory.

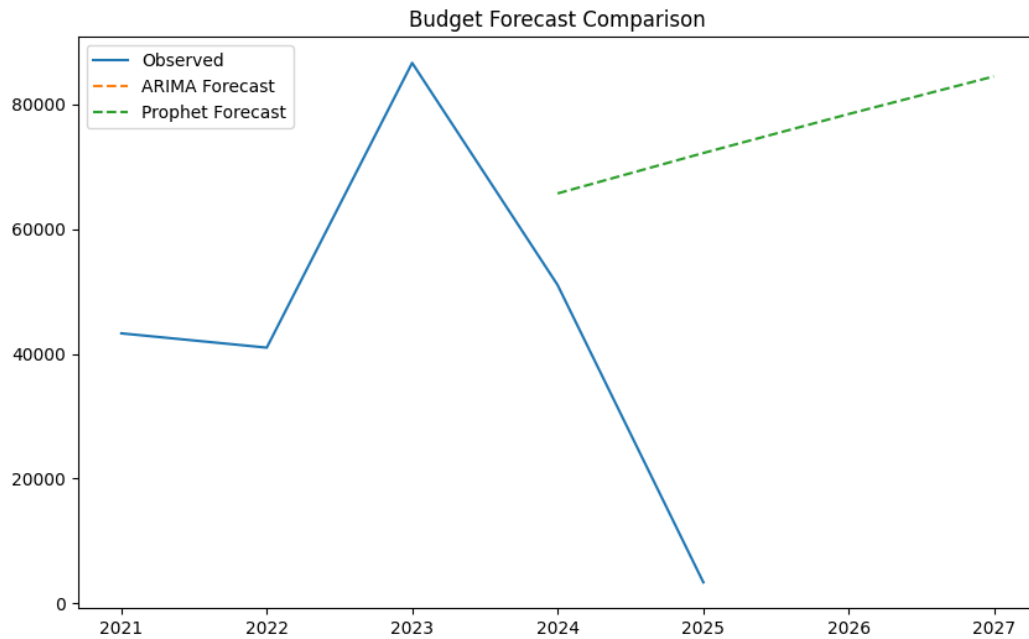


Figure 4. Time Series Forecast — Observed vs. Predicted (ARIMA and Prophet)

5.4 Deep Learning — LSTM

5.4.1 Objective

Beyond classical forecasting, a deep learning module was developed to explore the capacity of neural networks to capture complex, non-linear temporal dependencies in the data — patterns that may exceed the representational capacity of ARIMA or Prophet on richer, longer time series.

5.4.2 Methodology

Raw time series data was first normalized using Min-Max scaling to ensure stable gradient flow during training. Input sequences were constructed using a sliding window approach, transforming the data into the supervised learning format required by recurrent architectures. A **Long Short-Term Memory (LSTM)** network was built with multiple stacked layers and ReLU activation, trained using the Adam optimizer with Mean Squared Error (MSE) as the loss function. Predictions were subsequently inverse-transformed back to the original scale for interpretability.

5.4.3 Results and Comparison

The LSTM architecture demonstrated its capacity to track complex trends and capture long-range temporal dependencies that linear models cannot represent. On datasets with sufficient historical depth and rich behavioral patterns, LSTM can outperform both ARIMA and Prophet. Observed vs. Forecast visualizations (Figure 5) showed that the LSTM adapted more closely to fluctuations in the data, though on short series the classical models remain more stable due to their lower parameter count and built-in inductive biases. This comparison underscores the value of a multi-model strategy: classical models provide interpretability and robustness on limited data, while deep learning offers superior flexibility as data volume grows.

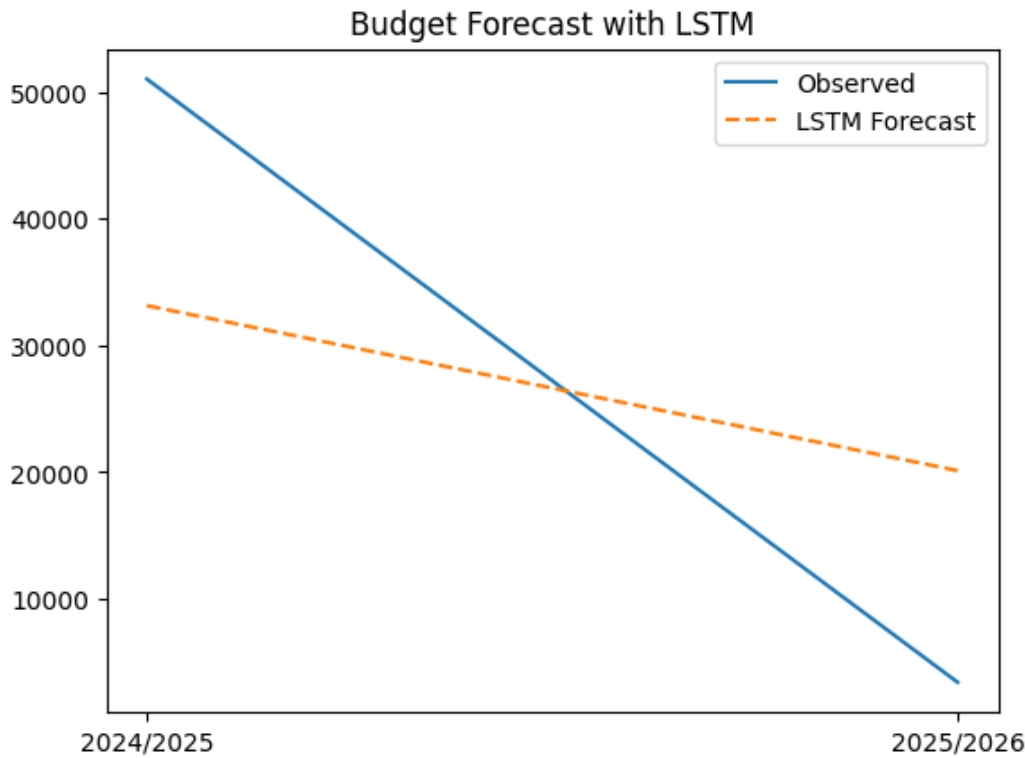


Figure 5. Deep Learning Forecast — Observed vs. Predicted (LSTM)

5.5 Clustering

5.5.1 Objective

The clustering module aimed to segment scout units into homogeneous groups based on their organizational characteristics, enabling leadership to identify distinct unit profiles, target support interventions, and compare performance across peer clusters.

5.5.2 Models Implemented

- **K-Means:** Partitions units into K clusters by minimizing intra-cluster inertia. Its simplicity and interpretability make it a strong baseline for structured tabular data.
- **DBSCAN:** Groups points based on local density and is capable of identifying clusters of arbitrary shape while explicitly labeling low-density points as noise — valuable for detecting outlier units.
- **Gaussian Mixture Model (GMM):** A probabilistic framework assigning each unit a soft membership probability across all clusters. This provides a richer representation of ambiguous cases at the boundary between performance profiles.

5.5.3 Evaluation

The optimal number of clusters for K-Means was determined using the **Elbow Method**, which showed a pronounced inflection at $K = 5$ (Figure 6). All models were further evaluated using:

- **Silhouette Score:** measures the separation between clusters relative to internal cohesion; higher values indicate better-defined clusters.

- **Davies-Bouldin Index:** measures cluster compactness and inter-cluster separation; lower values indicate superior clustering quality.
- **Pseudo-R²:** quantifies the proportion of total variance explained by the cluster structure.

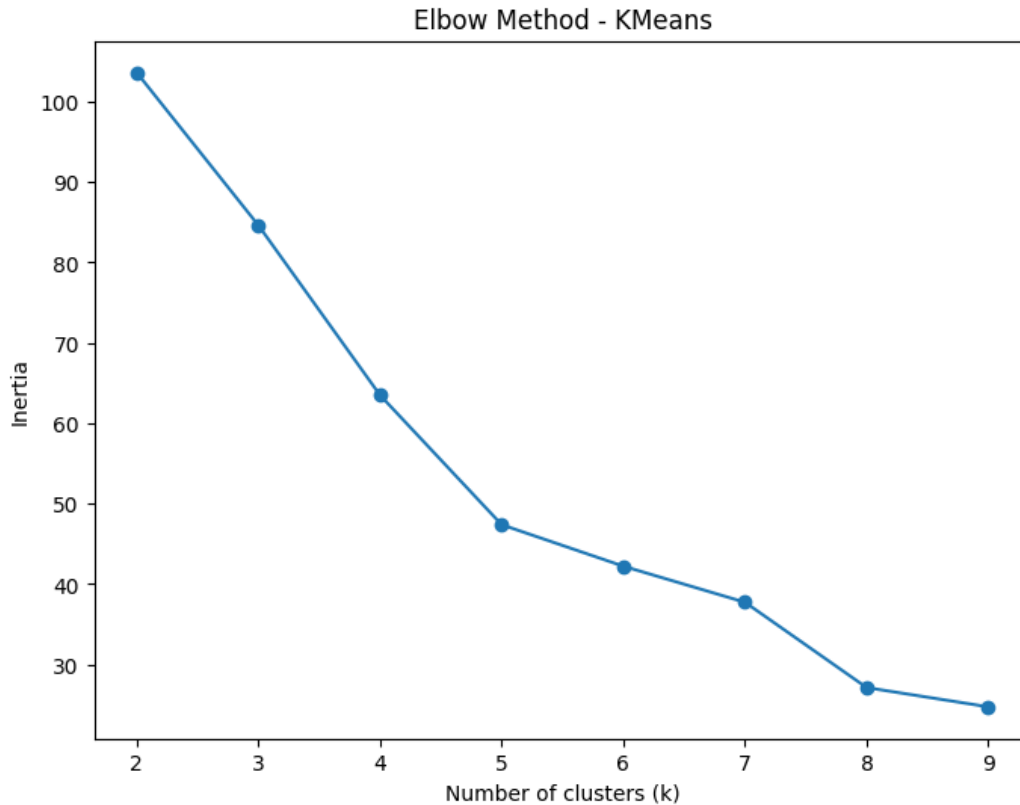


Figure 6. K-Means Elbow Method — Optimal Cluster Selection

5.5.4 Results

K-Means ($K = 3$) delivered well-separated, balanced clusters with strong evaluation scores. GMM produced comparable cluster assignments with the added benefit of probabilistic membership scores. DBSCAN proved more sensitive to its hyperparameters (`eps`, `min_samples`) and produced fewer consistently valid clusters on this dataset, though it remained useful for identifying noisy or atypical units. The consensus across metrics established K-Means and GMM as the most reliable approaches for this use case.

5.5.5 Interpretation and Visualization

PCA-based 2D projections (Figure 9) provided clear visual confirmation of cluster separation for both K-Means and GMM. Heatmaps of cluster profiles (Figure 10) enabled a direct comparative reading of mean values across key variables — `nb_members`, `nb_leaders`, `taux_fidelisation`, and `taux_evolution` — for each cluster. This profiling revealed three interpretable unit archetypes:

- **Cluster 1 — High-Fidelity:** strong member retention, stable leadership
- **Cluster 2 — Growth-Oriented:** positive evolution trends, expanding head-

count

- **Cluster 3 — At-Risk:** low engagement, declining membership, reduced leadership ratio

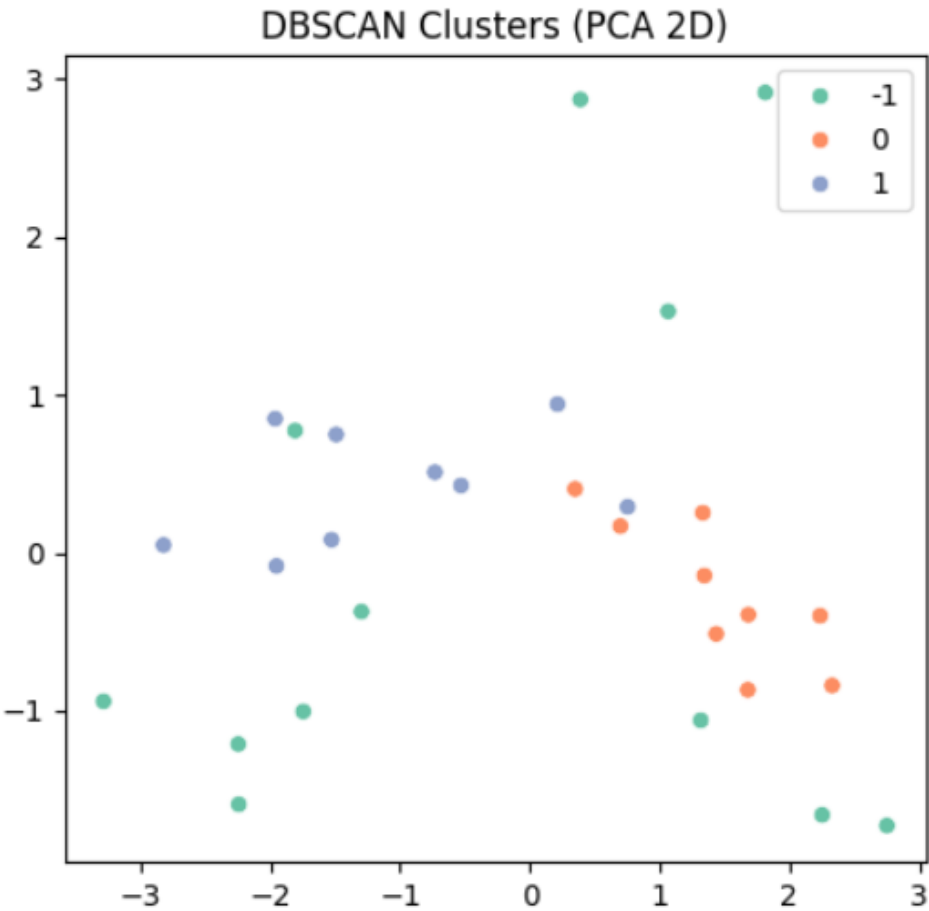


Figure 8. PCA 2D Visualization of Clustering Results (DBSCAN)

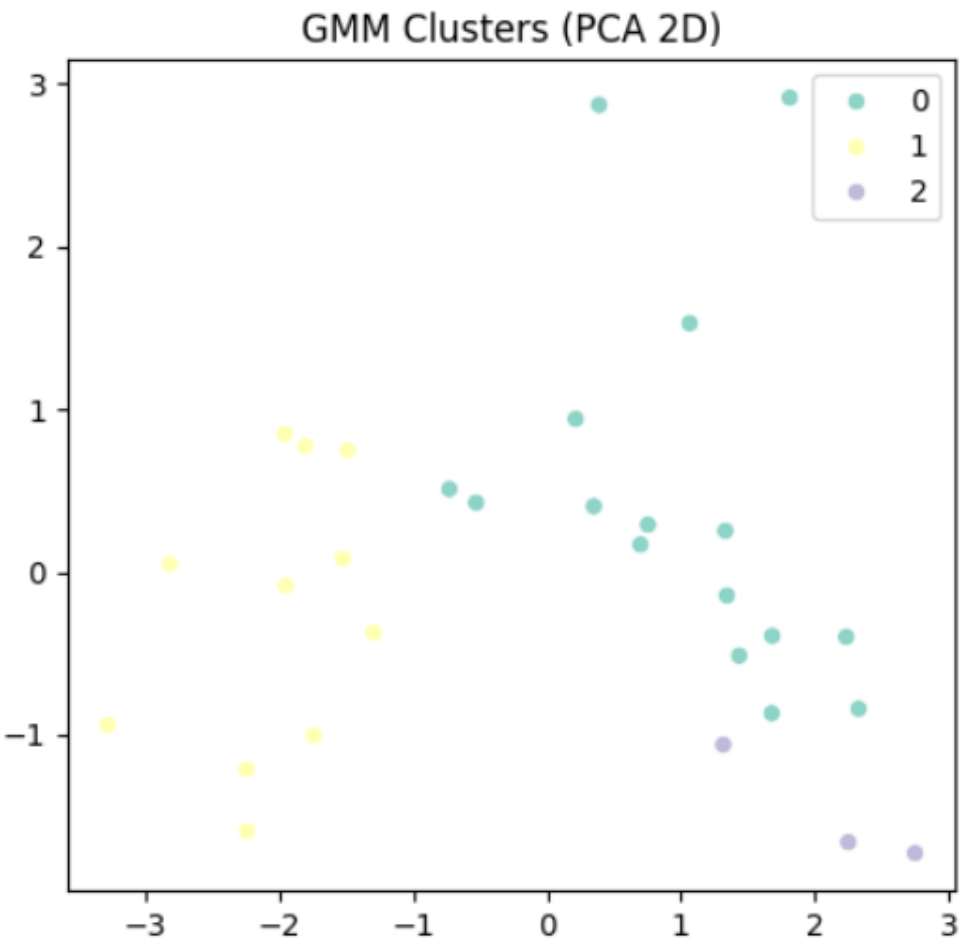


Figure 9. PCA 2D Visualization of Clustering Results (GMM)

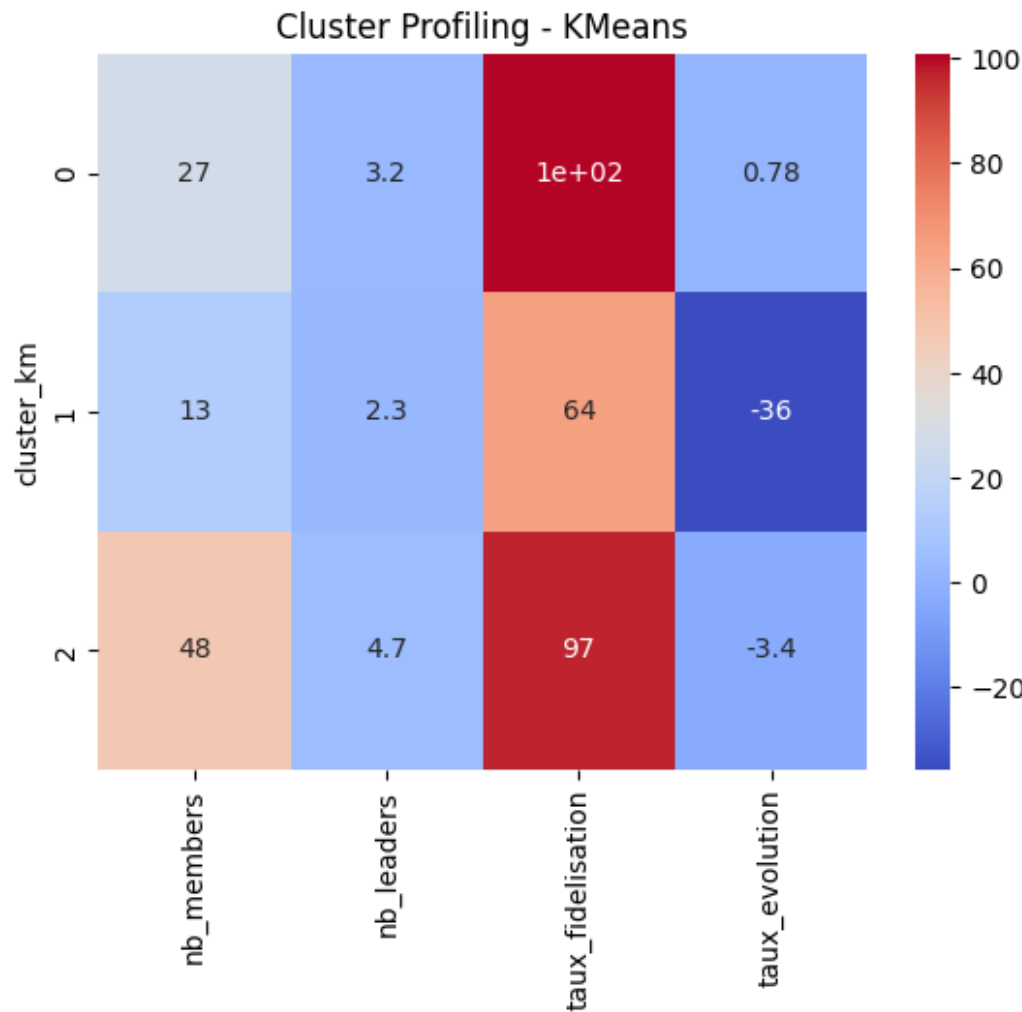


Figure 10. Cluster Profiling Heatmap — Mean Variable Values per Cluster

5.6 Regression — Membership Prediction

5.6.1 Objective

The regression module was designed to predict a continuous target variable — specifically the number of members per unit — using historical organizational features. Three models of increasing complexity were implemented, validated through cross-validation, and evaluated using standard regression metrics.

5.6.2 Models Implemented

- **Linear Regression:** The interpretable baseline, providing direct coefficient-level insight into the directional influence of each feature on membership counts.
- **Ridge Regression:** Introduces L2 regularization to penalize large coefficients, reducing overfitting on correlated features and producing more stable predictions — particularly relevant given the multicollinearity between organizational variables such as loyalty rate and evolution rate.
- **Random Forest Regression:** A non-linear ensemble approach aggregating predictions from multiple decision trees to capture complex interactions between features that linear models cannot represent.

5.6.3 Validation

All three models were validated using **5-Fold Cross-Validation**. Cross-validated R^2 scores were:

- Linear Regression: $R^2 \approx 0.62$
- Ridge Regression: $R^2 \approx 0.62$
- Random Forest: $R^2 \approx 0.66$

5.6.4 Results on the Test Set

Table 6. Regression Model Performance on Test Set

Model	R^2	MSE	RMSE	MAE
Linear Re- gression	0.71	33.1	5.75	4.38
Ridge Re- gression	0.81	22.6	4.75	3.91
Random Forest	0.87	15.5	3.94	3.47

Random Forest outperformed both linear models across all four metrics, achieving $R^2 = 0.87$ on the test set. Ridge Regression offered a meaningful improvement over the linear baseline, demonstrating that regularization alone captures a substantial share of the data complexity.

5.6.5 Interpretation

Residual analysis confirmed that Random Forest produced the smallest and most uniformly distributed prediction errors, with no systematic bias detectable across the value range. Feature importance rankings from Random Forest identified `taux_fidelisation` and `taux_evolution` as the dominant predictors. Actual vs. Predicted plots (Figure 11) visually confirmed that Random Forest tracked real values most closely.

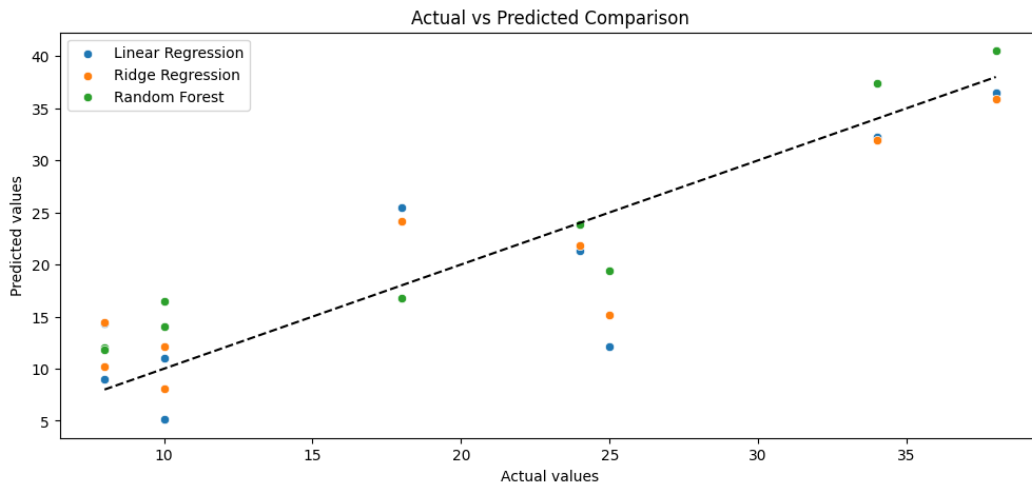


Figure 11. Regression — Actual vs. Predicted Membership Counts

5.6.6 Conclusion

Linear Regression established an interpretable baseline confirming the overall directional logic of the feature set. Ridge Regression improved robustness by controlling coefficient magnitude. Random Forest delivered the strongest performance ($R^2 = 0.87$) and was selected as the primary model for deployment, with Ridge Regression retained as a complementary explainability tool.

5.7 AI — Decision Support & Chatbot

Advanced AI capabilities were implemented on top of the ML layer:

- **Decision Support System:** Automated recommendations generated from analytical results — priority actions, resource allocation suggestions, and risk signals
- **Scenario Simulation:** Prospective scenarios allowing leadership to evaluate strategic choices before implementation
- **Weak Signal Detection:** Identification of early-warning patterns for organizational or financial risk
- **AI Chatbot:** Natural language interface built using LMFlow allowing leaders to query dashboards, request summaries, and receive recommendations in conversational form

Conclusion

In this chapter, we implemented and evaluated multiple machine learning and artificial intelligence models, including recommendation systems, anomaly detection, forecasting, clustering, regression, and deep learning approaches.

The obtained results demonstrated the effectiveness of the proposed analytical solutions and provided valuable insights for organizational decision-making. In the next chapter, we will evaluate the global performance, reliability, and quality of the developed system.

6 Phase 5 — Evaluation

Phase 5

Evaluation

Introduction

This chapter focuses on the evaluation phase of the project. We assess the performance of the implemented machine learning models, validate the quality of the analytical outputs, and conduct a global audit of the system architecture, security mechanisms, and infrastructure stability.

The purpose of this phase is to ensure that the developed platform satisfies both technical and business requirements before deployment.

6.1 Model Evaluation Summary

Table 7. ML Model Evaluation Approaches and Metrics

Model	Evaluation Approach & Metrics
Membership Forecast	Train/test split by season; RMSE, MAE, MAPE on held-out seasons
Budget Estimation	5-fold cross-validation; R^2 , RMSE, residual analysis
Participation Prediction	Stratified K-Fold; RMSE, R^2
Unit Clustering	Elbow method; Silhouette Score ≥ 0.60 target; Davies-Bouldin Index
Anomaly Detection	Precision/Recall on manually labeled anomalies; F1-Score
Recommendation System	Hit Rate @ K on past activity choices; Precision@K

6.2 System Audit

- A full system audit was conducted covering:
- **Data pipeline stability:** validation of Airflow DAG success rates and error log analysis
 - **Dashboard accuracy:** cross-verification of Power BI KPI values against raw

DWH queries

- **Model drift assessment:** monitoring of prediction accuracy degradation over time
- **Security audit:** review of authentication flows (password + facial recognition)
- **Performance audit:** Grafana dashboard review of API response times and infrastructure load

Conclusion

In this chapter, we evaluated the performance and reliability of the implemented models and conducted a complete system audit covering infrastructure, dashboards, APIs, and security mechanisms.

The obtained evaluation results confirmed the robustness and effectiveness of the proposed solution. In the next chapter, we will present the deployment architecture and the technologies used to operationalize the platform.

7 Phase 6 — Deployment

Phase 6

Deployment

Introduction

In this chapter, we present the deployment phase of the project. We describe the global system architecture, the integration of the different technologies, the deployment of the web application, the MLOps lifecycle management process, and the monitoring infrastructure.

This phase transforms the analytical solution into a fully operational intelligent platform accessible to all stakeholders of the scout organization.

7.1 Architecture Overview

Table 8. Deployed Architecture — Components and Technologies

Component	Technology & Role
ETL Pipeline	Talend Open Studio — Extract, Transform, Load from staging to DWH
Orchestration	Apache Airflow — DAG-based scheduling and monitoring of all data jobs
Data Warehouse	Star-schema relational DWH — central analytical store
BI Dashboards	Power BI — bilingual (FR/EN) interactive reports embedded in Angular
Web Application	Angular — unified frontend portal integrating dashboards and ML features
ML API	Flask REST API — serves predictions from trained ML models
MLOps	LMFlow — model versioning, training management, and life-cycle tracking
Monitoring	Grafana — real-time infrastructure and pipeline performance monitoring
Authentication	Password-based login + Facial Recognition (OpenCV / Deep-Face)
Chatbot	LMFlow-based conversational AI — natural language decision support

7.2 Web Application

The Angular web application serves as the unified entry point for all stakeholders. Key features include:

- **Dual authentication:** standard username/password login and facial recognition-based biometric login
- **Role-based access control:** differentiated views and permissions for Group Leadership, Unit Leaders, and Administrative Staff
- **Embedded Power BI dashboards** served directly within the Angular interface
- **Real-time ML prediction panels:** membership forecasts, unit risk indicators, anomaly alerts
- **Bilingual interface:** full French and English language support across all dashboards
- **Integrated AI chatbot** widget available on all authenticated pages

7.3 Application Web — Interface Utilisateur

7.3.1 Page d'Accueil — Sélection du Rôle

La page d'accueil de l'application constitue le point d'entrée unique pour tous les utilisateurs. Elle présente trois rôles distincts accessibles via une interface visuelle immersive : **Chef d'Unité**, **Trésorier**, et **Chef de Groupe**. L'application supporte trois langues : **Arabe (AR)**, **Français (FR)**, et **Anglais (EN)**, avec une bascule linguistique disponible en permanence dans la barre de navigation.

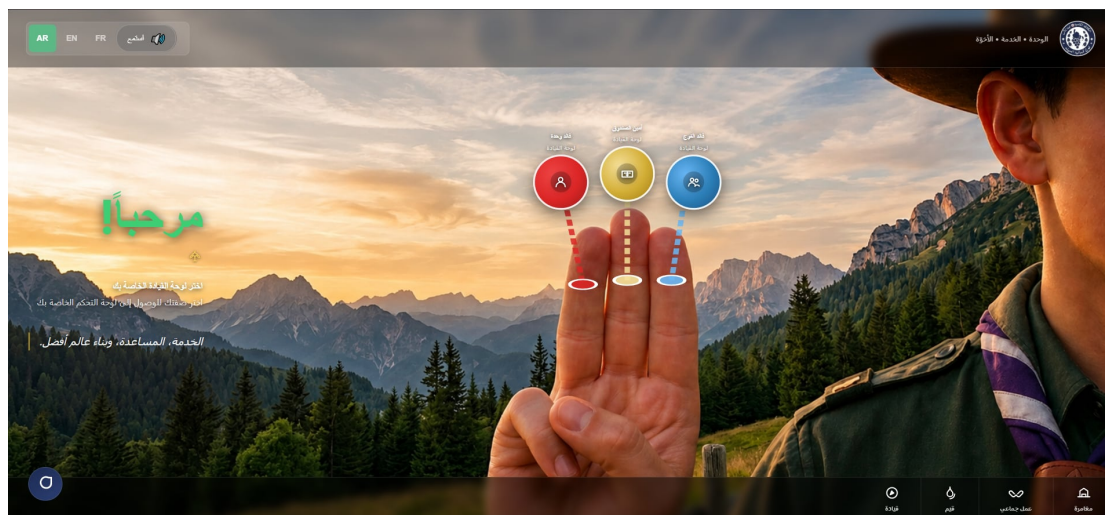


Figure 12. Page d'accueil — Sélection du rôle (version Arabe)

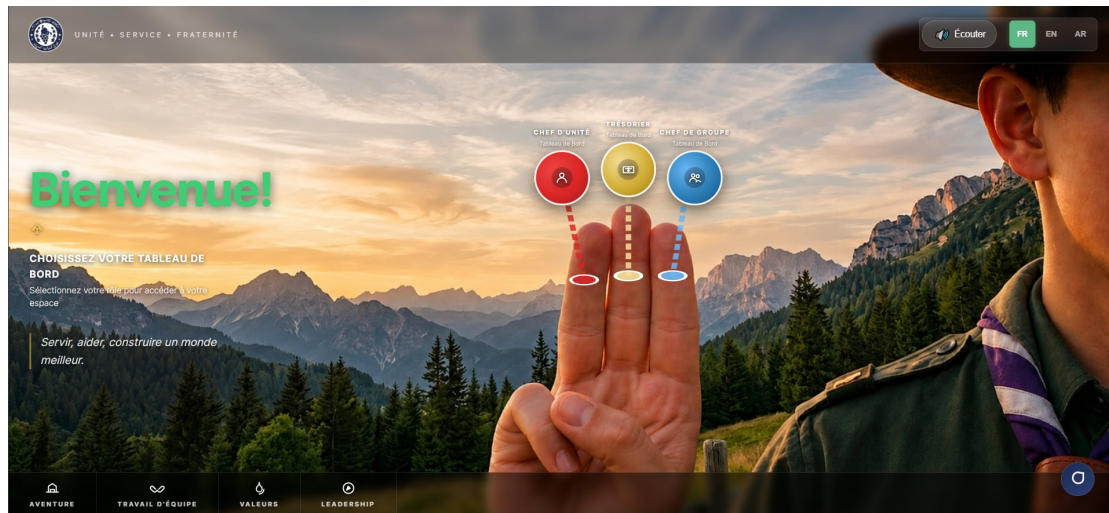


Figure 13. Page d'accueil — Sélection du rôle (version Française)

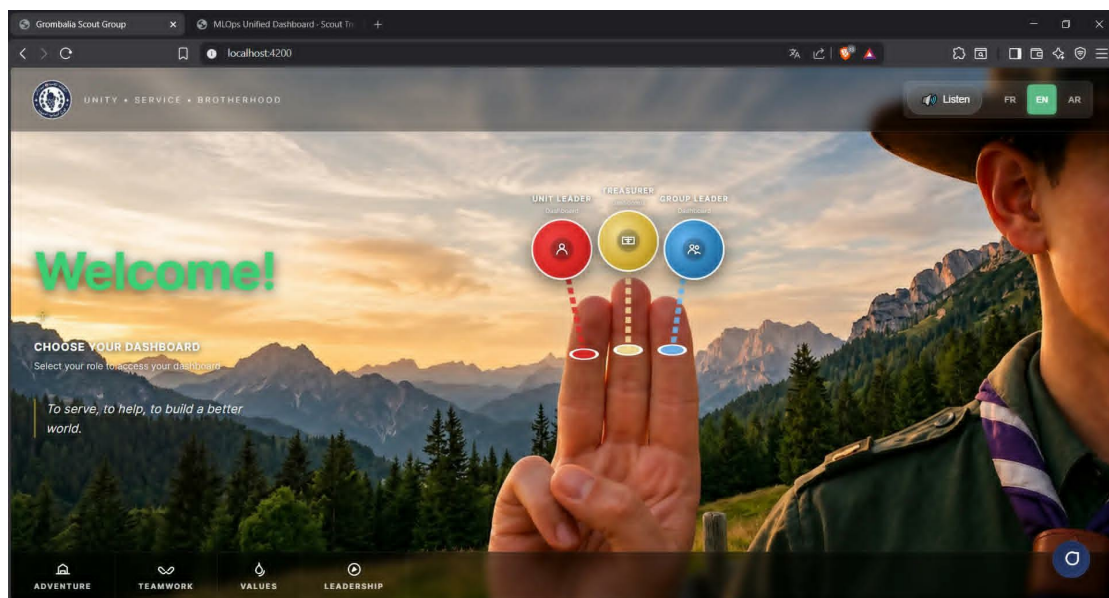


Figure 14. Page d'accueil — Sélection du rôle (version Anglaise)

7.3.2 Authentification — Pages de Connexion par Rôle

Chaque rôle dispose d'une page de connexion dédiée, proposant deux méthodes d'authentification :

- **Authentification par mot de passe** : saisie classique du nom d'utilisateur et du mot de passe
- **Authentification biométrique (Face ID)** : reconnaissance faciale via OpenCV / DeepFace

La différenciation visuelle entre les rôles est assurée par un code couleur distinct : **Rouge** pour le Chef d'Unité, **Bleu marine** pour le Chef de Groupe, et **Doré** pour le Trésorier.

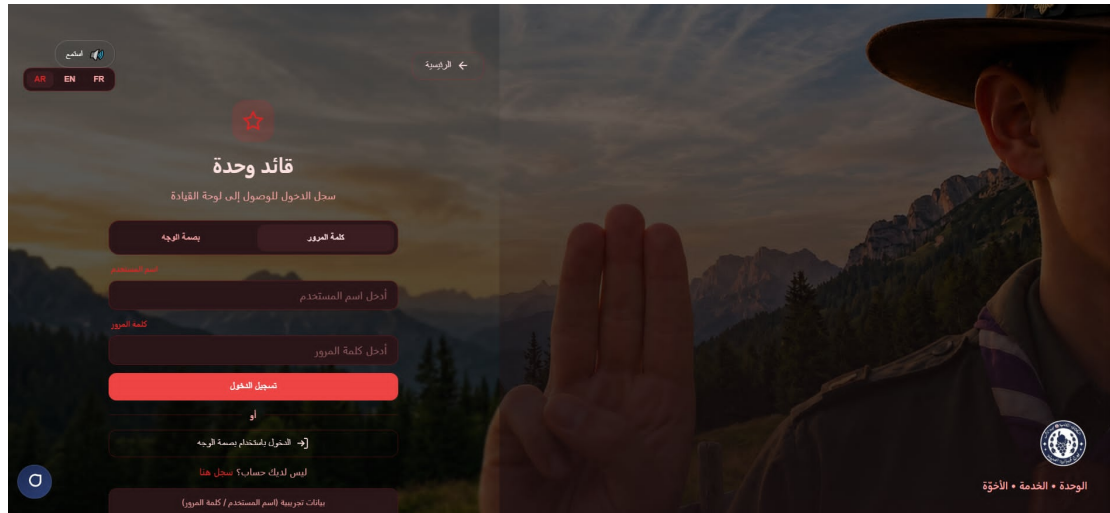


Figure 15. Connexion Chef d'Unité (AR)

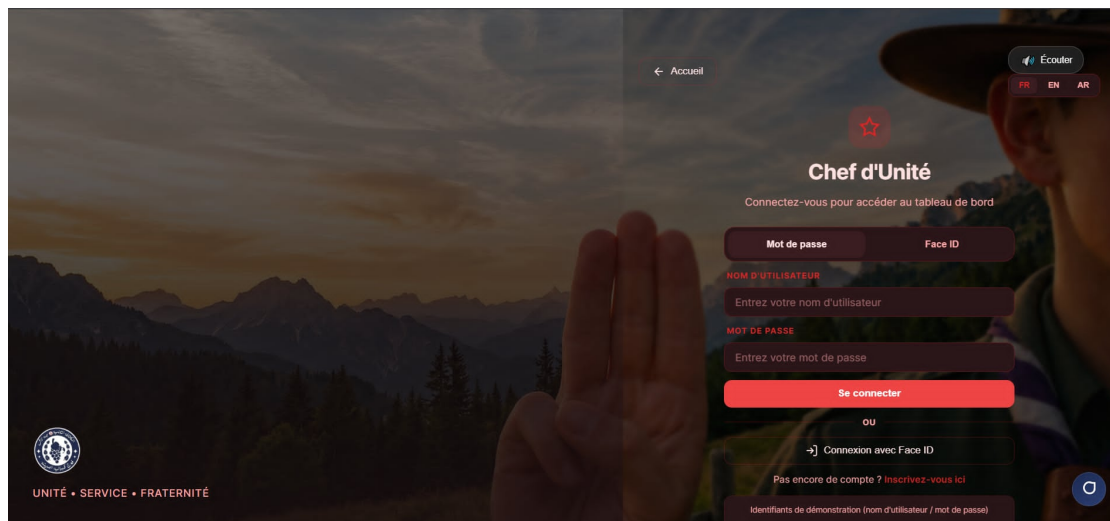


Figure 16. Connexion Chef d'Unité (FR)

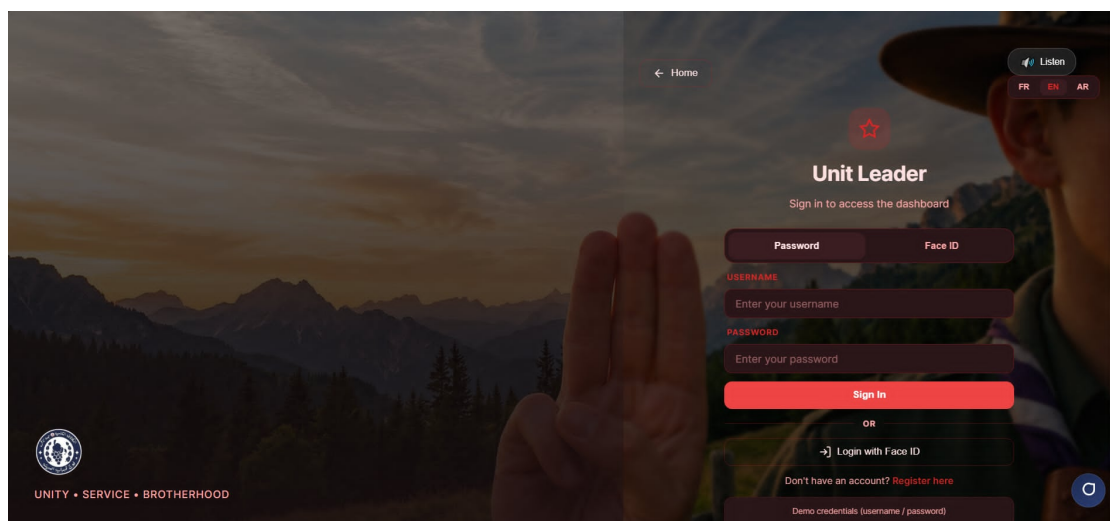


Figure 17. Connexion Chef d'Unité (EN)

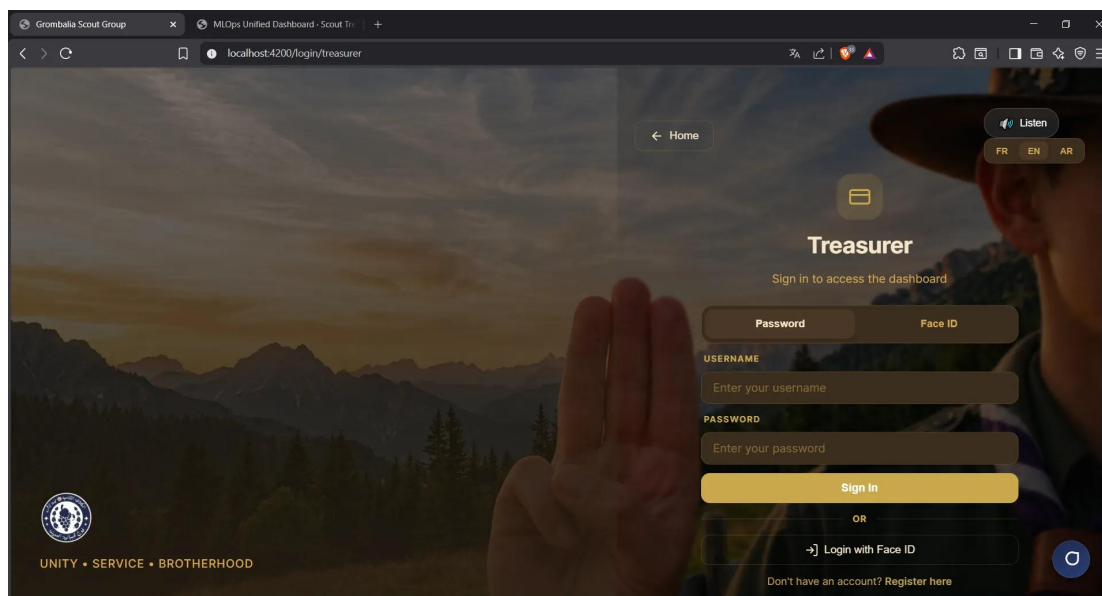


Figure 18. Connexion Chef de Groupe (EN)

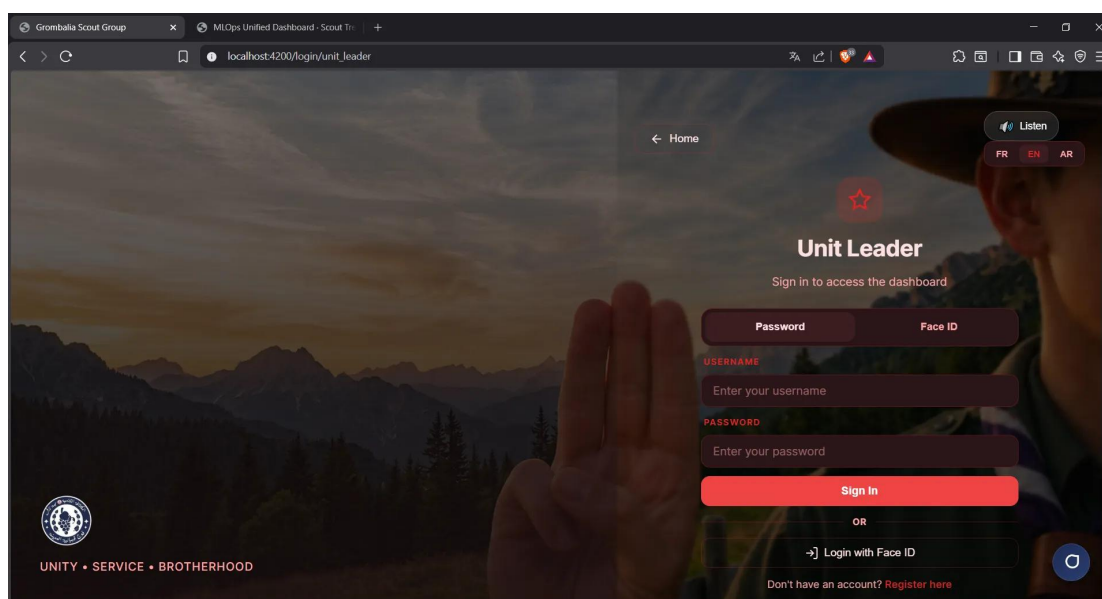


Figure 19. Connexion Trésorier (EN)

7.3.3 Tableau de Bord — Chef d'Unité

Le tableau de bord du Chef d'Unité intègre des rapports **Power BI** embarqués directement dans l'interface Angular. Il est structuré en trois vues complémentaires accessibles via des onglets : **Overview**, **Analysis**, et **Decision**.

Vue d'ensemble (Overview) La vue principale présente les indicateurs clés de performance de l'unité :

- 716 membres actifs au total
- 11.1% de taux de participation (cible : 70% — écart critique)

- **16.6%** de score de fidélisation
- **51.1%** d'indice de parité de genre (objectif ODD respecté)

Des graphiques interactifs illustrent la répartition des participants vs. total, la ventilation des présences par type d'activité, la localisation géographique des activités, et l'évolution des membres par unité et par saison.

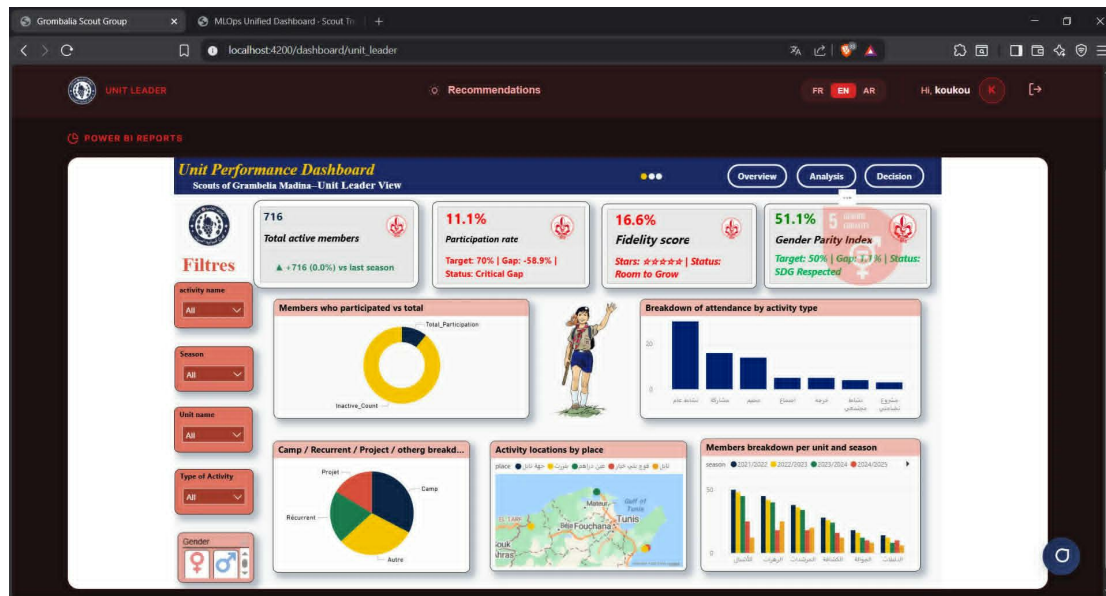


Figure 20. Tableau de bord Chef d'Unité — Vue d'ensemble (Overview)

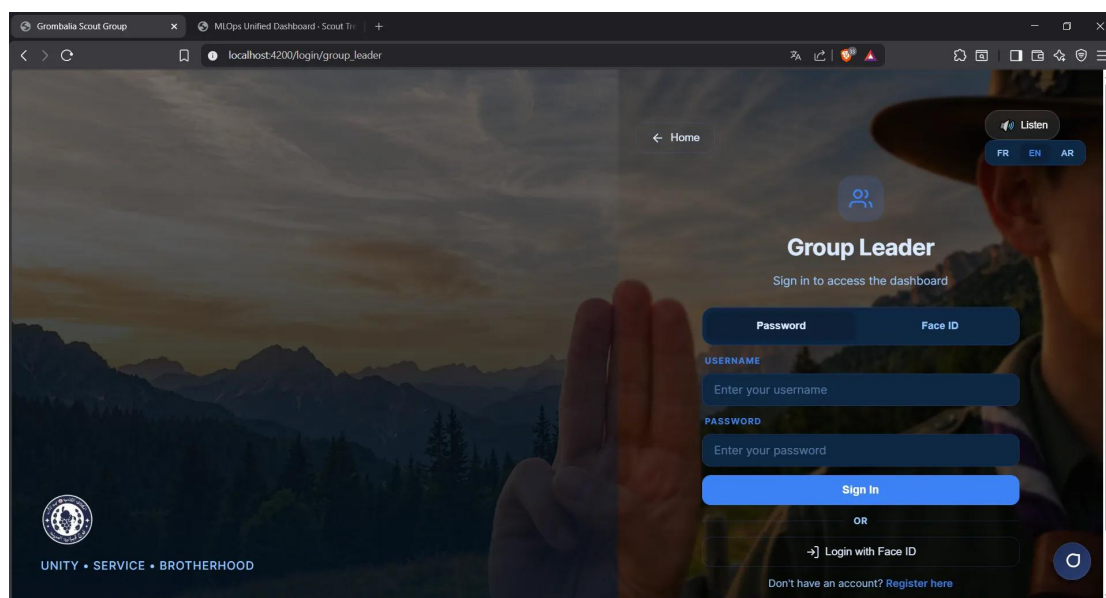


Figure 21. Tableau de bord Chef d'Unité — Vue d'ensemble complète avec filtres

Vue Analyse & Performance La vue analytique approfondit les métriques opérationnelles :

- **30** activités réalisées lors de la saison en cours

- 3 membres en moyenne par activité (cible : 15% — écart de 11.8%)
- 6 types d'unités actives
- 2025/2026 identifiée comme saison de pic de fidélisation

Elle intègre un module d'analyse des facteurs d'influence sur le niveau de participation (Key Influencers), un classement des activités par taux de participation, ainsi qu'une courbe d'évolution du taux de fidélisation sur les années.

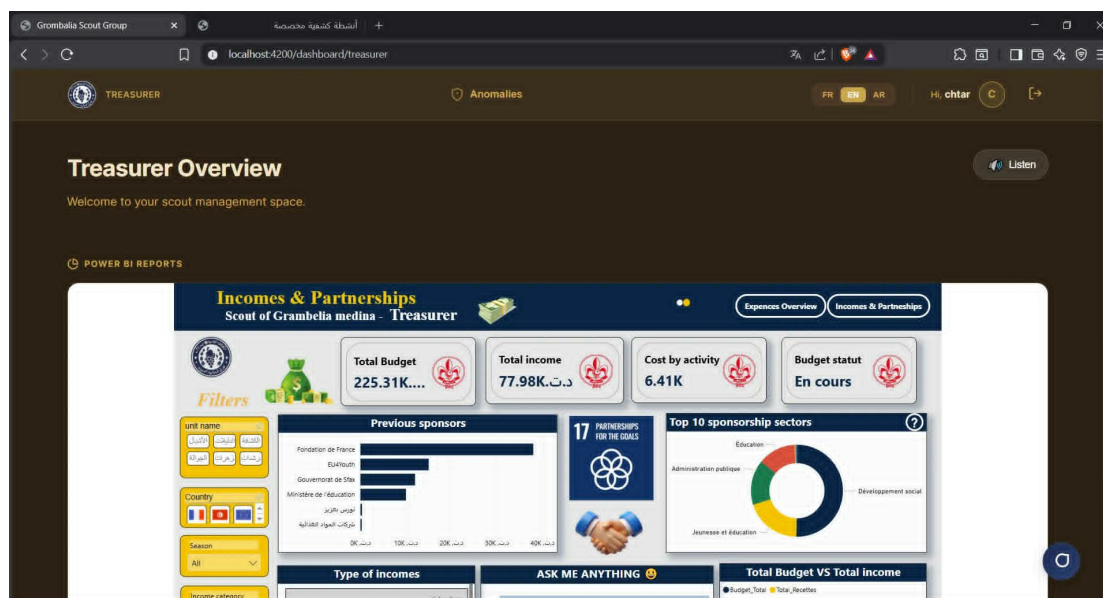


Figure 22. Tableau de bord Chef d'Unité — Analyse & Performance

7.3.4 Tableau de Bord — Trésorier

Le tableau de bord du Trésorier est organisé en deux rapports Power BI complémentaires : **Expenses Overview** et **Incomes & Partnerships**.

Vue Revenus & Partenariats (Incomes & Partnerships) Les indicateurs financiers globaux affichés incluent :

- **Budget total** : 225.31K DT
- **Total des revenus** : 77.98K DT
- **Coût moyen par activité** : 6.41K DT
- **Statut budgétaire** : En cours

Le tableau de bord présente également les principaux sponsors (Fondation de France, EU4Youth, Gouvernorat de Sfax, Ministère de l'Éducation), la répartition sectorielle des partenariats (Éducation, Développement social, Administration publique, Jeunesse), ainsi qu'un module *Ask Me Anything* permettant des requêtes en langage naturel sur les données.

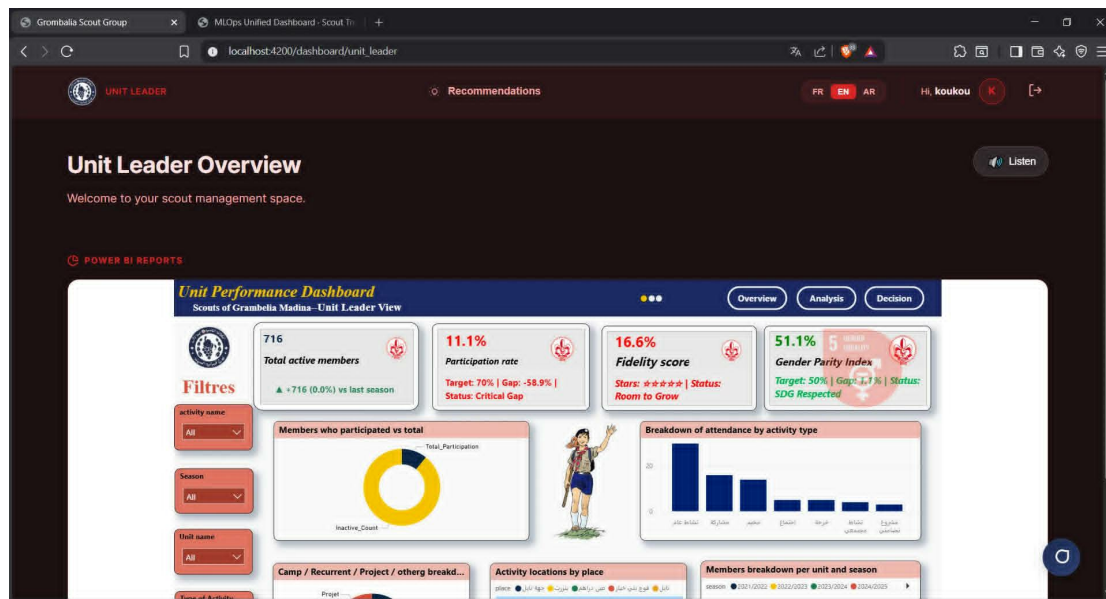


Figure 23. Tableau de bord Trésorier — Vue générale avec module Ask Me Anything

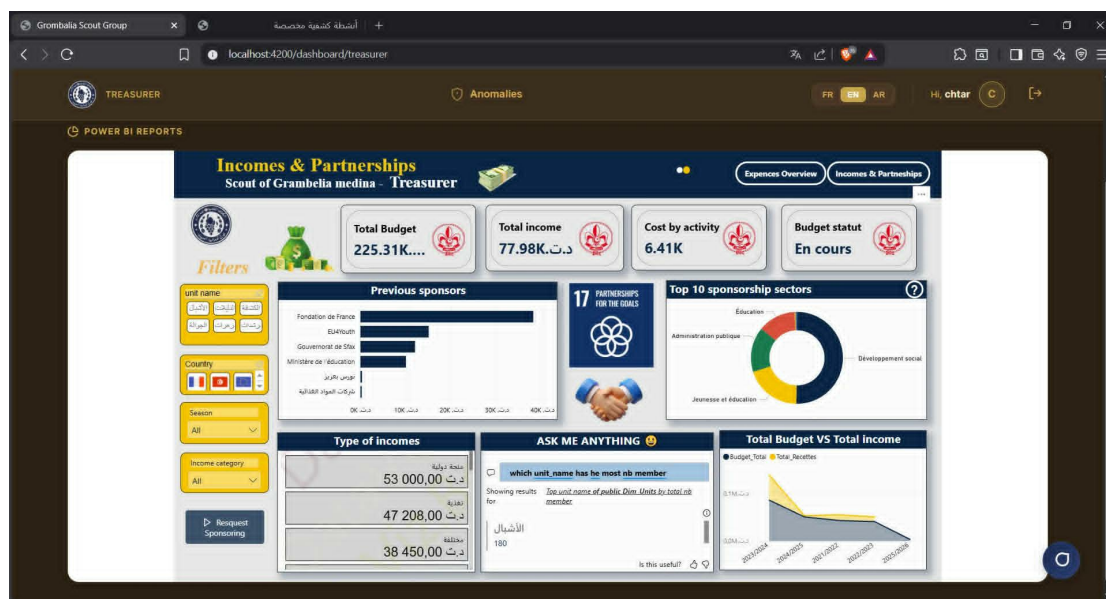


Figure 24. Tableau de bord Trésorier — Revenus & Partenariats

Vue Dépenses (Expenses Overview) La vue des dépenses offre une analyse complète des sorties financières :

- Total des dépenses : 147.33K DT
- Écart budgétaire : 77.98K DT
- Répartition des dépenses par unité sous forme de graphique circulaire
- Classement des 5 activités les plus coûteuses (dont le Camp d'Été 2022 : 38.45K DT)
- Évolution du budget par unité et par saison

- Localisation géographique des événements sur carte interactive

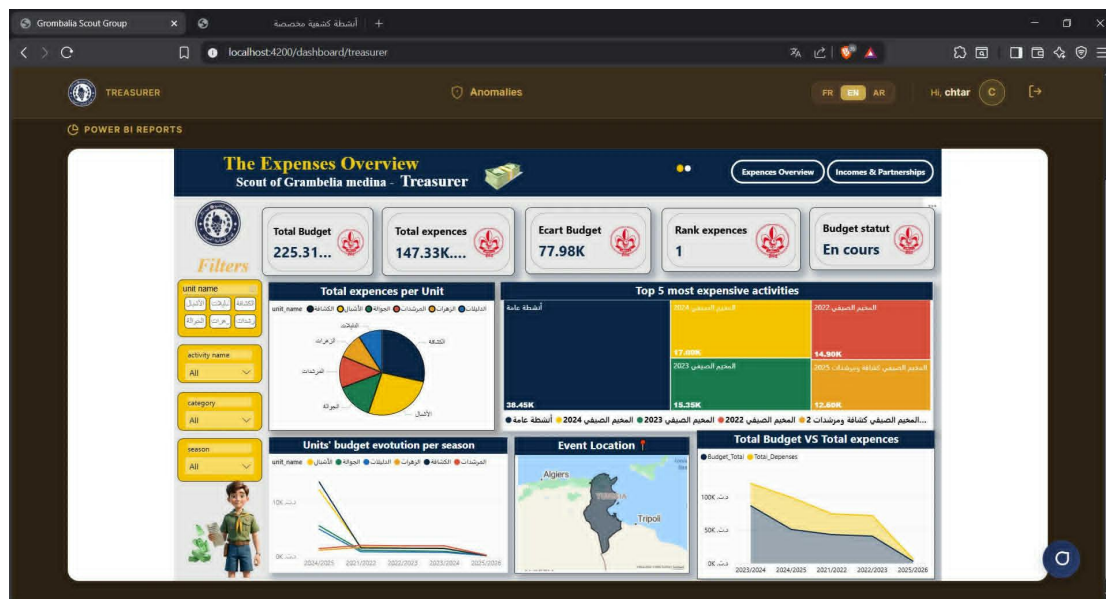


Figure 25. Tableau de bord Trésorier — Vue des Dépenses (Expenses Overview)

7.3.5 Système de Recommandation d'Activités

Un module de recommandation personnalisé a été déployé sous forme d'application distincte accessible depuis l'interface principale. Ce système permet à chaque membre de saisir son identifiant scout afin d'obtenir une liste d'activités recommandées générées par le modèle de filtrage collaboratif (SVD).

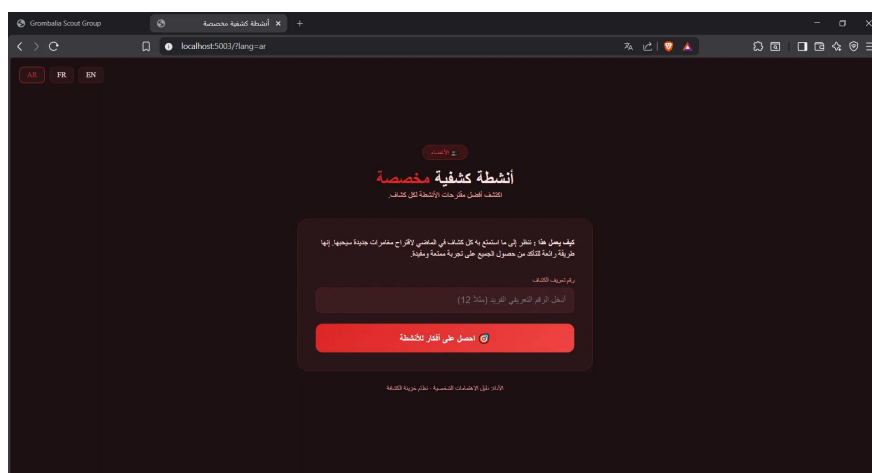


Figure 26. Interface du Système de Recommandation d'Activités (version Arabe)

7.3.6 Outils d'Aide à la Décision — Modules Complémentaires

Trois modules intelligents supplémentaires ont été déployés en parallèle de l'application principale, accessibles depuis l'interface et adressant des besoins opérationnels spécifiques.

Planification des Activités à Venir Ce module permet aux responsables d'estimer le nombre de scouts qui participeront aux prochaines activités, en se basant sur l'analyse des historiques de présence par saison et par unité. Il fournit ainsi une aide concrète à la logistique (quantités de ravitaillement, équipements nécessaires) en amont de chaque événement.

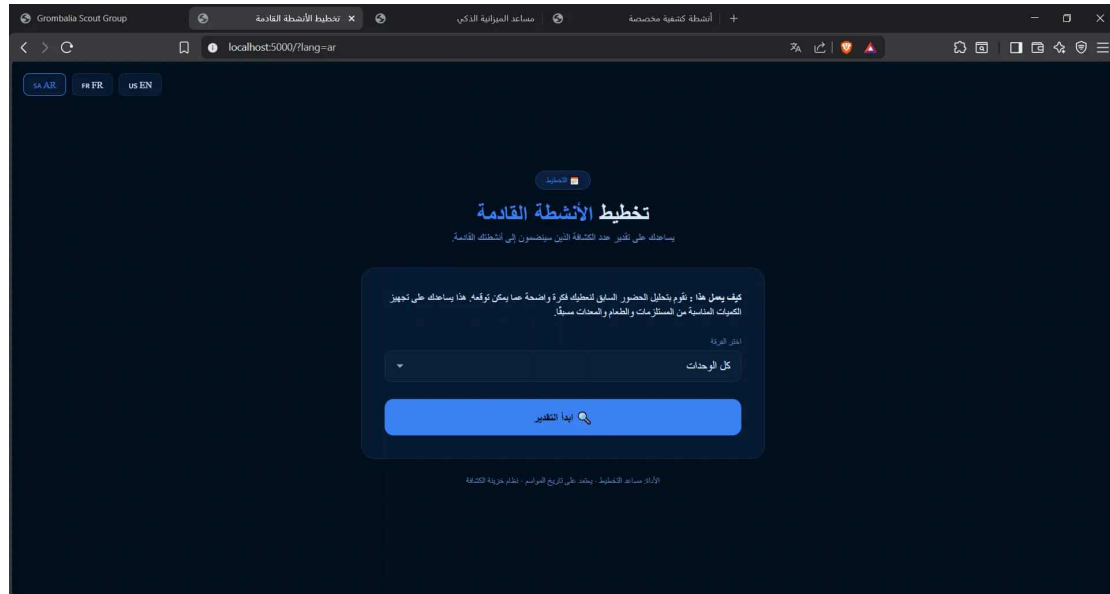


Figure 27. Module de Planification des Activités à Venir — Estimation de la participation (AR)

Group Growth & Belonging — Santé de l'Unité Ce module d'analyse de la croissance et de l'appartenance permet aux chefs d'unité d'évaluer la vitalité organisationnelle de leur groupe. En saisissant le nombre de scouts, de responsables et le taux de croissance récent, le système estime si les membres développent un fort sentiment d'appartenance à l'unité. Un indicateur de référence est fourni : une unité saine maintient au moins 9 membres actifs sur 10 à chaque saison.

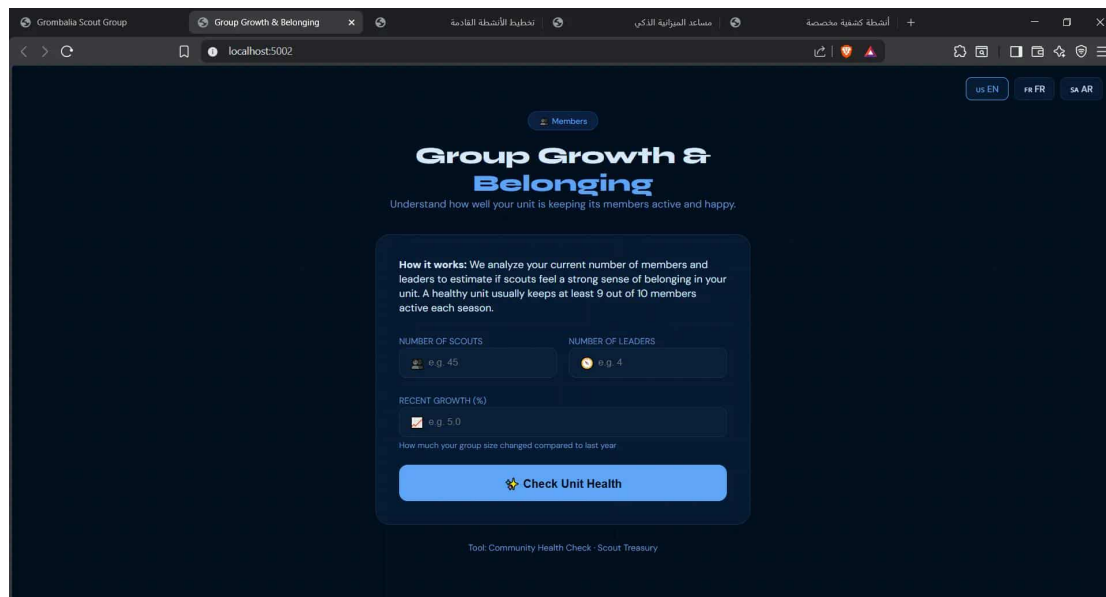


Figure 28. Module Group Growth & Belonging — Diagnostic de santé de l'unité (EN)

Scout Assistant — Chatbot de Navigation et de Support Un assistant conversationnel IA a été intégré directement dans l'interface principale de l'application. Accessible depuis toutes les pages authentifiées via un widget flottant, le **Scout Assistant** guide les utilisateurs en langage naturel vers les espaces appropriés selon leur rôle. À titre d'exemple, l'assistant propose :

- 1. Group Leader — Planification & Prévisions
- 2. Treasurer — Budget & Finance
- 3. Unit Leader — Activités & Membres

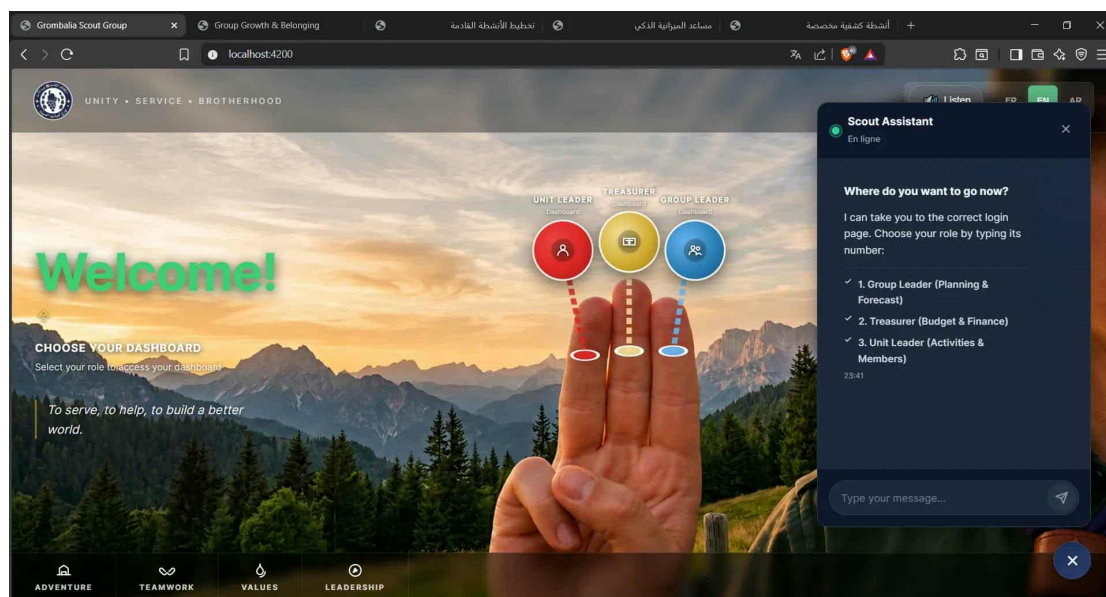


Figure 29. Scout Assistant — Chatbot IA de navigation et de support décisionnel

7.3.7 Synthèse des Fonctionnalités de l'Application

Table 9. Récapitulatif des fonctionnalités de l'application web par rôle

Rôle	Fonctionnalités Disponibles
Chef d'Unité	Dashboard Power BI (Overview / Analysis / Decision), module de recommandation d'activités, alertes de participation, authentification mot de passe + Face ID
Trésorier	Dashboard financier (Dépenses / Revenus & Partenariats), module Ask Me Anything, détection d'anomalies budgétaires, authentification mot de passe + Face ID
Chef de Groupe	Vue stratégique consolidée multi-unités, indicateurs de performance globaux, accès aux projections ML, authentification mot de passe + Face ID
Toutes interfaces	Support trilingue AR / FR / EN, chatbot IA intégré, lecture audio (accessibilité), navigation responsive

7.4 MLOps and Workflow Automation — LMFlow & n8n

Model lifecycle management was implemented using **LMFlow** combined with **n8n** workflow automation in order to ensure efficient orchestration of machine learning operations and intelligent automation processes.

The implemented MLOps architecture provides the following capabilities:

- **Model versioning:** all trained models are tracked with metadata, training date, dataset version, and evaluation metrics
- **Automated retraining workflows:** n8n workflows automatically trigger re-training pipelines when model performance decreases below predefined thresholds
- **Workflow orchestration:** n8n coordinates interactions between data pipelines, APIs, monitoring tools, and ML services
- **Model registry management:** centralized storage and management of production-ready models
- **Deployment automation:** validated models are automatically promoted to production environments
- **Alerting and notifications:** automated notifications generated in case of pipeline failures, anomalies, or infrastructure issues

The integration of n8n significantly improved the automation and maintainability of the ML lifecycle while reducing manual operational tasks.

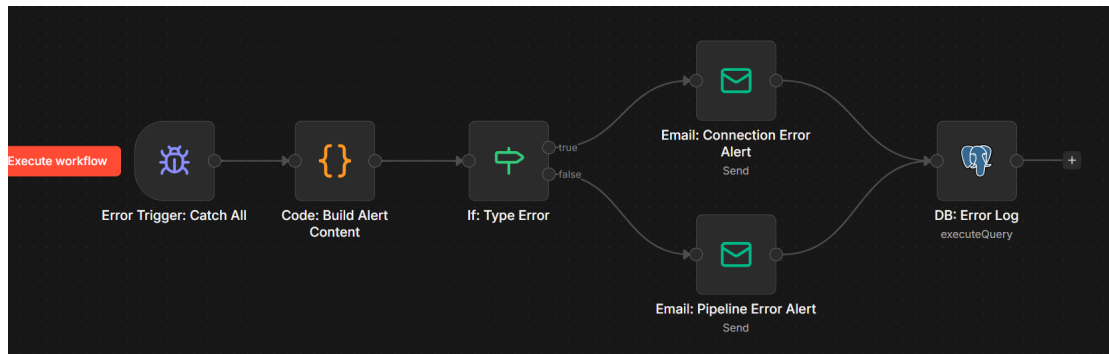


Figure 30. n8n Workflow Automation for MLOps Processes

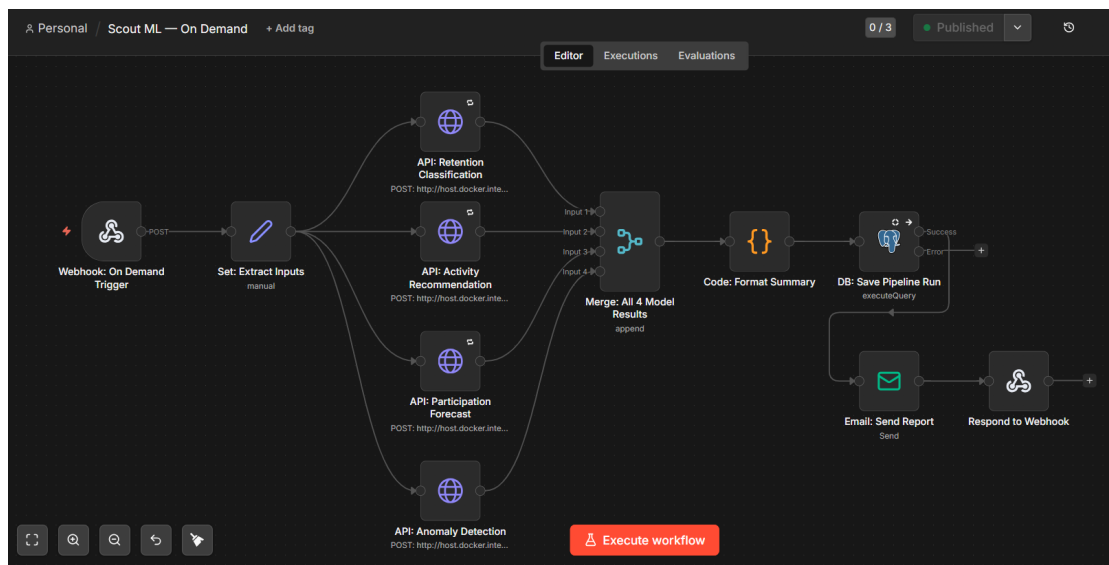


Figure 31. n8n Workflow Automation for MLOps Processes

7.5 Monitoring Infrastructure — Prometheus and Grafana

A complete monitoring infrastructure was deployed using **Prometheus** and **Grafana** in order to ensure real-time supervision of the platform and maintain high system reliability.

Prometheus was responsible for collecting and storing metrics generated by the different services of the architecture, including:

- API response times
- CPU and memory consumption
- Airflow pipeline execution metrics
- Data warehouse activity statistics
- Machine learning model serving performance

The collected metrics were then visualized through interactive dashboards built with **Grafana**. These dashboards provide real-time visibility into system health, infrastructure load, pipeline stability, and anomaly alerts.

This monitoring layer enables proactive maintenance, rapid issue detection, and performance optimization across the entire platform infrastructure.

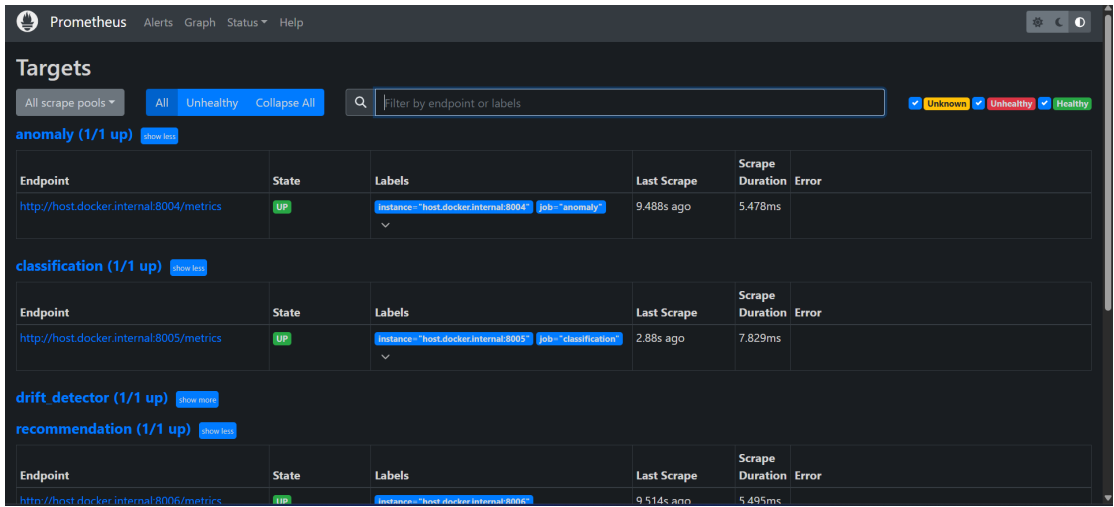


Figure 32. Grafana Real-Time Monitoring Dashboard

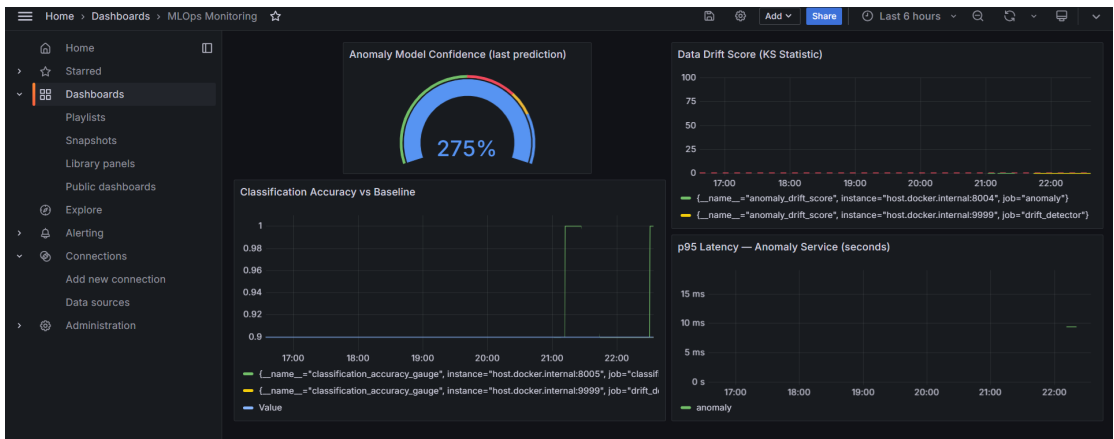


Figure 33. Grafana Monitoring Dashboard

Conclusion

In this chapter, we presented the deployment architecture and the technologies used to operationalize the platform. We detailed the integration between the frontend application, the data warehouse, the machine learning APIs, the MLOps infrastructure, and the monitoring tools.

The deployed architecture ensures scalability, maintainability, and real-time accessibility of the analytical services. The next section summarizes the complete technical stack used throughout the project.

8 Technical Stack Summary

Introduction

This section summarizes the complete technological ecosystem used throughout the project lifecycle. It presents the different tools, frameworks, platforms, and programming technologies involved in data ingestion, processing, machine learning, deployment, monitoring, and application development.

The diversity of the selected technologies reflects the multidisciplinary nature of the project and ensures the robustness of the proposed solution.

Table 10. Technical Stack by Layer

Layer	Technologies Used
Data Ingestion & ETL	Talend Open Studio for Data Integration
Web Scraping	Python (BeautifulSoup, Scrapy)
Pipeline Orchestration	Apache Airflow
Data Warehouse	Relational DBMS — Star Schema
Business Intelligence	Microsoft Power BI (bilingual FR/EN dashboards)
Machine Learning	Scikit-learn, XGBoost, LightGBM, Prophet, TensorFlow/Keras
MLOps & LLM	LMFlow
Backend API	Python Flask (REST API)
Frontend Application	Angular (SPA integrating BI + ML)
Authentication	JWT (password-based) + OpenCV/DeepFace (facial recognition)
Monitoring	Grafana
Conversational AI	LMFlow-based chatbot
Version Control	Git / GitHub

Conclusion

In this section, we summarized the complete technical stack adopted for the implementation of the project.

The selected technologies enabled the construction of a scalable, intelligent, and fully integrated BI/ML/AI platform adapted to the needs of the scout organization.

The combination of Business Intelligence, Machine Learning, MLOps, and modern

web technologies ensured both analytical performance and operational usability.

9 Conclusion

This project successfully delivered a comprehensive, end-to-end BI/ML/AI platform for the Scout Group of Grombelia Al-Medina, transforming fragmented administrative records into a structured, intelligent decision-support ecosystem. Following the CRISP-DM methodology ensured that every technical deliverable was grounded in clear business objectives validated with stakeholders at each phase transition.

The solution goes well beyond traditional Business Intelligence: machine learning models provide predictive capability, the recommendation system actively supports operational planning, and the AI chatbot democratizes data access for non-technical leadership. The deployment of MLOps practices through LMFlow ensures the platform remains accurate and maintainable as new seasonal data accumulates.

The multi-model strategy adopted across all ML tasks — comparing classical and modern approaches, linear and non-linear methods, parametric and density-based clustering — reflects a rigorous analytical philosophy that prioritizes both performance and interpretability. The integration of deep learning (LSTM) alongside classical forecasting (ARIMA, Prophet) positions the platform to scale gracefully as the volume and richness of historical data grows over successive scout seasons.

Future enhancements may include expansion of the web scraping pipeline to cover federation-wide activity data, integration of real-time membership intake forms, and development of a mobile companion application for unit leaders in the field.

End of Report — BISMART 4ERP-BI2 • 2025/2026